

Was There Informed Trading in LIBOR Referenced Interest Rate Futures?

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Abstract

During 2008 a series of allegations emerged in respect of systematic attempts to manipulate the London Interbank Offered Rate (LIBOR). In this paper we implement a version of the probability of informed trading measure first introduced in Easley et al. (1996), for the Eurodollar futures market over the 1996 to 2013 period. The objective of this paper is an ex-post review of the effectiveness of the PIN in determining changes in the information structure of the market around documented episodes of recorded manipulation of the benchmark rate, from the various publicly available regulatory reports. In keeping with previous studies on interest rate derivatives, we find that the average PIN is far higher than for the equity market at or around 2/3 to 3/4. When implementing a rolling measure of the PIN to detect time variation, we find a clear pattern of increases and decreases in informed trading relative to the recorded activity in the current regulatory reports. Furthermore, we also find a strong maturity effect that appears to arise from the strong time variation in trading in these contracts over their life-cycle, from inception to maturity. We construct a series of experiments to determine the significance of the trading time for different epochs from 1996 to 2014 using all inside quotes and trades. Our results indicate that the PIN could have been used as an early warning of unusual activity in the LIBOR reference rate and anecdotally we demonstrate that on specific dates identified by the CFTC and FSA the recursively estimated PIN reaches a peak over 1,000 basis points higher for certain near-delivery contracts than for others further away in the forward curve.

Keywords: Microstructure, Informed Trading, PIN, LIBOR, Eurodollar, Market Manipulation

1. Introduction

The London InterBank Offered Rate (LIBOR) and the European InterBank Offered Rate (EURIBOR) act as a reference rate for the majority of interest rate derivatives traded in global markets. However, the probity of this rate-setting mechanism has been called into

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question and several cases of manipulation of the LIBOR fix have been recorded since 2008. This paper exploits new information released by the various regulatory cases and uses it to conduct a field test of a commonly used method to assess the degree of asymmetric information from the order flows of traded assets, in this case LIBOR referenced Eurodollar futures. The Eurodollar futures market is referenced to the three-month dollar LIBOR rate (henceforth 3M-LIBOR) for maturities out to ten years. We utilize a large sample of trade and quote data to estimate the probability of informed trading (PIN) using the model of Easley et al. (1996). The results paint a mixed picture of the effectiveness of the PIN for interest rate futures. In most of the documented cases the PIN varies from the long-run averages in a statistically distinctive manner; however, other effects, such as high-frequency quoting around maturity dates, are found to generate noise that has a similar impact on the PIN.

The background of our experiment lies in the run-up to the 2007 financial turmoil and anomalies in the LIBOR rate in the immediate aftermath. An analysis conducted by the Wall Street Journal (WSJ) on May 29, 2008 suggested that banks may have reported flawed interest data for 3M-LIBOR¹. The LIBOR is set by a panel of banks that report their borrowing costs for various maturities and a censored mean is then taken that eliminates the top four and bottom four of the distribution leaving an average of the middle eight banks (see Table 2). Since no transactions may have been made that day for many maturities, an estimate of the rate is offered to fill in the blanks. The WSJ article indicated that the banks may have been suppressing their borrowing costs to conceal signs of financial distress, and subsequent investigation by global regulators found that the bank has indeed the case during the period from 2008 to 2010. The investigation also uncovered a systematic manipulation of the rates pre-2007 and indicated that the motivation for this activity was to assist the parts of the banks that traded in interest rate derivatives. This pre-crisis period will be the focus of the paper, since the documented events provide direct evidence of heterogeneous

¹The WSJ analysis suggests banks may have reported flawed interest data for LIBOR. (C., Mollenkamp and M., Whitehouse, Study Casts Doubt on Key Rate, <http://online.wsj.com/news/articles/SB12120070376202713>) [Last accessed August 19, 2014]

Table 1: Number of requests by derivatives traders to their LIBOR fixing desks for the 3M-USD LIBOR and other LIBOR or similar rates.

LIBOR Currencies	Barclays	Lloyds	RBS	UBS	Rabobank	Total
1M-USD LIBOR	111	1	7	22	19	160
3M-USD LIBOR	127	36	34	57	49	303
Sterling LIBOR	22	60	21	36	63	202
Euribor	5	1	46	54	105	211
YEN LIBOR	160	53	3	81	179	476
EuroYen	-	-	-	86	-	86
Total	425	151	111	336	415	1,438

The comparison of the number of requests between 3M-USD LIBOR and other LIBOR currencies mentioned in CFTC documents in the matter of LIBOR manipulation for Barclays, RBS, UBS, Rabobank and Lloyds Bank.

information across traders in specific markets such as derivative market. According to the current set of regulatory documents released, as of July 2014, most of the requests for fix adjustments by derivatives traders to various bank rate setters directly focus on the 3M-LIBOR rate. While many derivatives contracts are closely associated with this rate, the largest by far in terms of notional outstanding, is the futures contract on a three-month Eurodollar contract (see Table 1).

This paper provides evidence for the effectiveness of the PIN model in actually detecting informed behaviour in the Eurodollar futures market. To accomplish this we collect all trades and inside quotes for Eurodollar futures from 1996 to 2014, a data set in excess of two billion observations. We believe this to be the largest study of its type ever conducted. We compute the PIN using a variety of approaches and construct test groups to see whether the PIN varies systematically with the events suggested in the regulatory filings, versus a set of controls. In general, our control groups show average PIN variation around “normal” events in the futures contract life-cycle – in this case the maturity of the contract.

The reports on LIBOR manipulation for each bank provide us with a unique field experiment to test the PIN as the number of events is very large and the specificity of the objectives of those involved is very clearly stated. It is worth reviewing the rationale behind this approach. The PIN seeks to detect trades that appear to be the result of informed signals prior to some pay-off event. For equities the future dividend is often somewhat neb-

ulous at higher frequencies, however in futures markets it is not. Each day the outstanding contracts are marked to market using the last one minute of trading or the mid-price of the final outstanding quotes. At maturity the contracts are settled to cash based on the price of 100 minus the 3M-LIBOR on the third Wednesday of the settlement month. It should be stressed that for all “manipulation” is supposed to take place in the reference rate (the 3M-LIBOR) and not directly in the Eurodollar itself.

Our foundational assumption is that given that a particular bank is able to influence the reference rate and hence possibly influence the mark to market and certainly influence the final settlement price, then the traders of this bank will have more complete information on the final payoff of this asset over both the trading days and over the overall maturity of the contract. This informational advantage should then be reflected in the PIN, relative to days when no manipulation of the reference rate took place. We will illustrate using examples and summary statistics from the regulatory reports that requests from derivatives traders tended to take place with relatively short notice periods of between one day and one week. The degree of variation in the reference rate that can be created by “high or low balling” the rate is relatively small. However, we also illustrate that the size of the notional positions in the Eurodollar market is so large that there is still a significant incentive to shave the reference rate in your favour.

Overall, our findings are mixed. We find that the PIN reacts strongly to certain types of events and the variation is statistically significant relative to our control group. However, the PIN is always very high for Eurodollar futures market, PIN is averaging over 0.5 for the majority of our sample, compared to the equity market where observed PIN, for which the PIN usually varies between 0.1 and 0.6 from a survey of prior studies. We also find that the PIN is not stable relative to the various estimation assumptions. However, because this is almost entirely driven by the instability of the “trade classification” algorithm used to determine buying pressure. So, robustness check for trade classification is necessary, we applied volume weight average price for bid and ask with tick rule and compare with Lee&Ready algorithm. The results of PIN from these two classification are quite significant difference. Highlighting the difference pin is in agreement with Ke (2014) finding that the

difference algorithm which giving the difference number between buy and sell possibly has a great impact on the value of PIN. This also accords with Easley et al. (2012a) which the pin is difficulty to estimate in highly active market or in high frequency trading, so they proposed “Volume-synchronized Probability of Informed Trading (VPIN)” to solve these problems. However, we leave full investigation of these issue to future work.

This paper is organized as follows §(2) reviews the current research on detecting informed trading. §(3) provides some of literature reviews and a detailed background to interest rate derivatives including LIBOR and Eurodollar Futures trading on CME’s Globex. §(4) includes data sources, data pre-processing and data statistic descriptive. §(5) outlines the version of the PIN model used in this study. §(6) explores the PIN and presents the empirical results. Finally, §(7) provides some conclusions and outlines directions for future work in this area.

2. Related Work

2.1. The current LIBOR manipulation

The LIBOR manipulation was first literatures by Wall Street Journal Mollenkamp and Whitehouse (2008) and Financial Times (Mackenzie and Tett (2008a,b)) conjecture that banks have been understated true LIBOR submissions rate to appear less risky about funding problems. Accordingly, the first academic published paper was studied about alternative fixings for LIBOR interest rate benchmark. The early studies of LIBOR manipulation focused on the cost of borrowing as it jump in spreads between the Overnight Index Swaps (OIS) and the LIBOR (Taylor and Williams (2008)). The comparison research between LIBOR submissions to other competitive rate in conjunction with the risk of borrowing cost is continuing such as (Wong (2009),Abrantes-Metz et al. (2011),Monticini and Thornton (2013)).Kuo et al. (2012) also extend these studies into moderately informative about borrowing cost by using a variety range of other public data beyond the LIBOR. Snider and Youle (2010) illustrate suggestive evidence that LIBOR manipulation during financial crisis period of 2007 to 2008 were misunderstanding, they suggest that the manipulation

were under the concerned of banks trading profit linked to their portfolio instead of trying to appear more healthier.

The most impact study was presented by Abrantes-Metz et al. (2011) when they testing their hypothesis compare LIBOR with other short term borrowing rate and evaluate the banks quotes with individual CDS spreads. Despite this study does not intend to provide any evidence of the manipulation, the statistical in this paper somehow appear to be consistent with dishonestly behavior for LIBOR fixing. Furthermore, Abrantes-Metz et al. (2011), Rauch et al. (2013) implemented the Benford's law on its distribution to investigate the artificial submission rates on the LIBOR. The implement of regression analysis was presented by (Chen (2013)) when he developing equilibrium LIBOR submission strategy and quantifies the LIBOR bias. In the same way the latest studies by Fouquau and Spieser (2015) using a specific linear regression model to detect some significant breaks in the LIBOR data to investigate the possible of manipulation.

2.2. The Probability of Informed Trading

The study of market microstructure in the presence of information asymmetries in the financial securities market has received considerable attention from academics and practitioners. Initially, current studies of informed trading has been developing from Grossman and Stiglitz (1980), Hellwig (1980) Admati (1985), Hasbrouck (1991), Kyle (1985) proposed 'Kyle's Lambda' with sequential auctions. Also, the relationship between informed trading and return (Glosten and Milgrom (1985), Admati and Pfleiderer (1988)), and with trade size Easley and O'Hara (1987). Arriving orders from informed traders are updated in a probabilistic fashion were focused by Glosten and Milgrom (1985) Easley and O'Hara (1987) and Easley and O'Hara (1992) who developed a probability of informed trading model that has a nicknamed 'PIN' and has subsequently proved to be very popular because of its easy to explain the link between the existence of information (good news, bad news), trading time and security prices.

Specifications of PIN has been widely adopted to the economics of financial risk. This is related empirical works to address issues, as information content of the time between

trades (Easley et al. (1996)), trades size (Easley et al. (1997a)), analyst coverage (Easley et al. (1997b)). Electronic market order flow (Brown et al. (1999)), stock splits (Easley et al. (2001)), corporate investment decision (Vega (2006), Ascioglu et al. (2008), Chen et al. (2007), Brockman and Yan (2009)), market mechanism (Heidle and Huang (2002), Grammig et al. (2001)), asset pricing (Easley et al. (2002), Aslan et al. (2011), Aslan et al. (2011))

Studies of the market microstructure in the presence of the PIN that impacts the price of risky assets were undertaken by Kang (2010), Idier and Nardelli (2011) and Chen and Zhao (2012). This phenomenon is often referred to the PIN model developed by Easley et al. (1996) to analyze the role and impact of information-based trading on risky asset. This is confirmed in our empirical work where the PIN is used for the risk measure in financial asset in the presence of asymmetry information. However, critical issues in the PIN measure is become a major concern, several papers show that the PIN suffers several biases for different reasons such as trade classification (Boehmer et al. (2007)) and boundary solutions especially for high liquidity asset, that's why VPIN is one of the solution proposed by the same authors of PIN (Easley et al. (2011a, 2012b)). Advantages of VPIN: avoid computational problem and biased; capture the risk variations at intraday levels while the PIN model does not; however, we leave full investigation for future work.

In view of all that has been mentioned so far, one may suppose that the studies of market microstructure in the presence of informed trading by using a probability of informed trading approach is a stabilized measurement tool for asymmetry information. While the PIN approach is quite popular for securities trading, only a tiny portion of the literature is involved with the interest rate derivative market. Thus, the gap in these previous studies offers a great opportunity to test some of the current measures of informed trading such as the PIN approach of Easley et al. (1996) with a very liquid interest rate derivative market.

3. Background of Interest Rate Derivatives Market and LIBOR Manipulation

The purpose of this research is to investigate the comparative microstructure of the Eurodollar Future markets during the LIBOR manipulation event. As a useful natural

experiment, the incident of LIBOR manipulation offers a great opportunity for academics to impute and observe parameters of interest in an information asymmetry model and to study the development of those estimators incorporated with the information content.

To understand the Eurodollar futures based on 3-M US Dollar LIBOR, it is deemed to be useful to introduce the background of the London InterBank Offered Rate or the LIBOR, since this is the benchmark or key interest rate for Eurodollar futures. The original LIBOR was introduced in 1984, when the BBA or British Bankers' Association and other parties, such as the Bank of England, formed a standard settlement rate of contractual terms on interest rate swaps. Two years later, the BBA published the first LIBOR as an official interest rate for a variety of financial instruments transacted across the multi-international financial market, notably syndicated loans, forward rate agreements, interest rate futures, interest rate swaps, and interest rate options (Abrantes-Metz et al., 2012). The LIBOR submission is now administered by the NYSE: ICE LIBOR unit. At 11.00 am on each weekday, the panel banks submit their rates to an administrator. Then, the highest four and the lowest four of the quoted rates are removed and the remaining middle rates are used to calculate a simple arithmetic mean. Finally, at 11.30am LIBOR is quoted and announced by Thomson Reuters. Since September 2013 the ICE LIBOR unit makes the fixing available to various data vendors (including Thomson Reuters) for distribution to the market and for settlement on maturing interest rate derivatives set relative to this rate. Currently, the daily LIBOR rates are quoted on 10 major currencies, including the Australian Dollar, British Pound, Canadian Dollar, European Euro, Danish Kroner, Japanese Yen, New Zealand Dollar, Swedish Kroner, Swiss Franc and US Dollar.

In London, the LIBOR rate is based on the rates quoted by 16 banks selected by the BBA to provide a daily rate for LIBOR calculation. Currently in the United States, 18 banks are selected by the BBA to form a panel to construct the US Dollar LIBOR. Three US and 16 non-American banks are included in the panel for the US dollar fixing. The US banks are: Bank of America, JP Morgan Chase and Citibank. The non-US Banks are: Bank of Tokyo-Mitsubishi UFJ, Barclays Bank, BNP Paribas, Credit Agricole CIB, Credit Suisse, Deutsche Bank AG, HSBC, Lloyds TSB Bank, Rabobank, Royal Bank of Canada, Société

Générale, Sumitomo Mitsui Banking Corporation, The Norinchukin Bank, The Royal Bank of Scotland Group and UBS AG.

As a benchmark of interest rates, the LIBOR is widely used for various financial instruments including standard interbank products such as forward rate agreement, interest rate swaps, interest rate futures, mortgage rates, standard loan rates and Eurodollar Futures trading. Likewise, in the United States around 60 to 80% of prime adjustable rate mortgages and sub-prime mortgages were indexed to the US LIBOR. In addition, the American municipalities borrowed around 75% of their money through products that were linked to the LIBOR.

3.1. Eurodollar Futures

By definition, Eurodollars are US Dollar deposits located outside the legal jurisdiction of the United States. The majority of Eurodollar deposits are located in London, however the Eurodollar deposit account really has very little to do with the futures contract named after it. The Eurodollar future contract made its debut in December 1981 on the Chicago Mercantile Exchange (CME), and this was the first future contract ever settled to cash. Since the transaction market for the Eurodollar futures contract is not so well understood outside the interest rate derivative community, we will explain the specifics in some detail in this section.

The Eurodollar futures themselves are interest rate derivatives with reference to the 3M London interbank offered rate (LIBOR), that have a face value of \$1 million with a 3-month maturity and the futures contracts have a maturity of up to ten years. The regular contracts expire either in March, June, September or December, extending outwardly for 40 quarterly expiring contracts in the long term; shorter maturity monthly contracts are traded, however, their relative volume, trading activity and quoting activity are several orders of magnitude lower than the quarterly contracts. The last trading day is the second business day prior to the third Wednesday of the delivery month in both New York and London. At any given time, the exchange lists four of the monthly series, bringing the total number of available Eurodollar future contract maturity types to 44 – namely, 40 quarterly and 4 monthly serial

contracts.

Eurodollar futures are traded around the clock on Central Standard Time (CST; Chicago time) from Sunday to Friday, 23 hours a day. On a typical trading day, the CME is divided into two stages: the pre-opening trading stage and the bilateral stage. The Eurodollar futures are exchanged exclusively in the open outcry on regular CME floor trading between 7:20 am to 2:00 pm. In addition, trading on Globex solely begins at 5:00 pm CST on Sunday, which represents the beginning of Monday's trading, and continues to operate overnight until 4:00 pm CST on the following day. However, Globex closes for regularly scheduled maintenance between 4:00 pm and 5:00 pm CST every day. For instance, 5:00 pm on Sunday represents the beginning of the trading session for Monday's trading day and it continues until 4:00 pm on Friday, when Globex closes for the weekend and re-opens again at 5:00 pm on Sunday to begin the next week's trading. Currently, more than 70% of Eurodollar futures are traded simultaneously on the electronic trading platform; CME Globex platform,² since it offers more liquidity and has lower transaction costs. Eurodollar futures, both short and long, are used extensively to finance and/or replicate the floating leg of swap contracts and the notional outstanding in these contracts has grown in concert with the notional size of the swap market (indeed according to CME the aggregate size of outstanding Eurodollar futures is usually about 10% larger than the swap market. Swap contract floating rate rebalancing is commonly conducted at or near the quarterly international money market (IMM) dates that correspond to the delivery dates on the quarterly Eurodollar futures.

The Eurodollar future is an interest rate future contract that is particularly related to the LIBOR – therefore, the price is quoted to 100 (or the total price is equal to 100 less the implied 3M-US Dollar LIBOR interest rate). For instance, if the implied interest rate is 0.250%, the IMM index is quoted as 99.750 (IMM Index = $100.000 - 0.250\% = 99.750$), and this is also the price of a futures contract. The gain or loss of these contracts is described by one basis point (0.01%), which equates to a 25.00 US dollar movement in contract value, derived as follows: $\$1 \text{ million notional loan} \times (0.01\%) \times (90/360) = \25.00 . However, the

²CME Group, Understanding Eurodollar Futures, <http://www.cmegroup.com/trading/interest-rates/understanding-eurodollar-futures.html> [accessed August 18, 2014].

minimum price fluctuation is set at half a basis point or 0.005%, equal to \$12.50, based on \$1 million face value 90-day instruments. For example, if the price was rising from 96.99 at the beginning of the day and closed at 97.00 (implying a LIBOR decrease from 3.01 to 3.00%), \$25 is paid from the seller's margin account into that of the buyer so that buying the contract is equivalent to lending money, and selling the contract short is equivalent to borrowing it.

Individually, Eurodollar futures, in effect, parallel forward rate agreements (FRAs), but provide more liquidity by virtue of the clearing house and being marked to market. Of key importance is the cash settlement – Eurodollar traders will receive cash compensation on the same day rather than having to wait until the expiry date. Moreover, large institutions tend to use Eurodollar futures to manage their risk or portfolio to avoid interest rate risk. To hedge against this risk, borrowers will enter into short future contracts to avoid the risk of rising interest rates while lenders will enter into long future contracts to avoid the risk of falling interest rates. However, there are speculators who purely seek to make a profit from Eurodollar trading by betting on the change in interest rates (future directional). Institutional investors will use Eurodollar futures contracts to lock in an interest rate today for their borrowing or lending in the future. Eurodollar futures are the most widely traded short-term interest rate futures in the world, with average daily volume of more than 2 million. Futures contracts are the most active interest rate futures traded in the world, typically in the 7 to 9 million range in the shortest maturity futures.³

3.2. The Current LIBOR Manipulation Cases

Our analysis of informed trading in the Eurodollar Futures market in this paper relates to documents published by both the media and financial regulators. However, our core source will be the CFTC reports relating to the current group of banks and brokers that have settled a fines with various national regulatory bodies. As of August 25, 2014 this group consists of Barclays, UBS, RBS and Rabobank. Each bank has been investigated and

³CME Group, volume data, <http://investor.cmegroup.com/investor-relations/downloads.cfm> [accessed August 19, 2014].

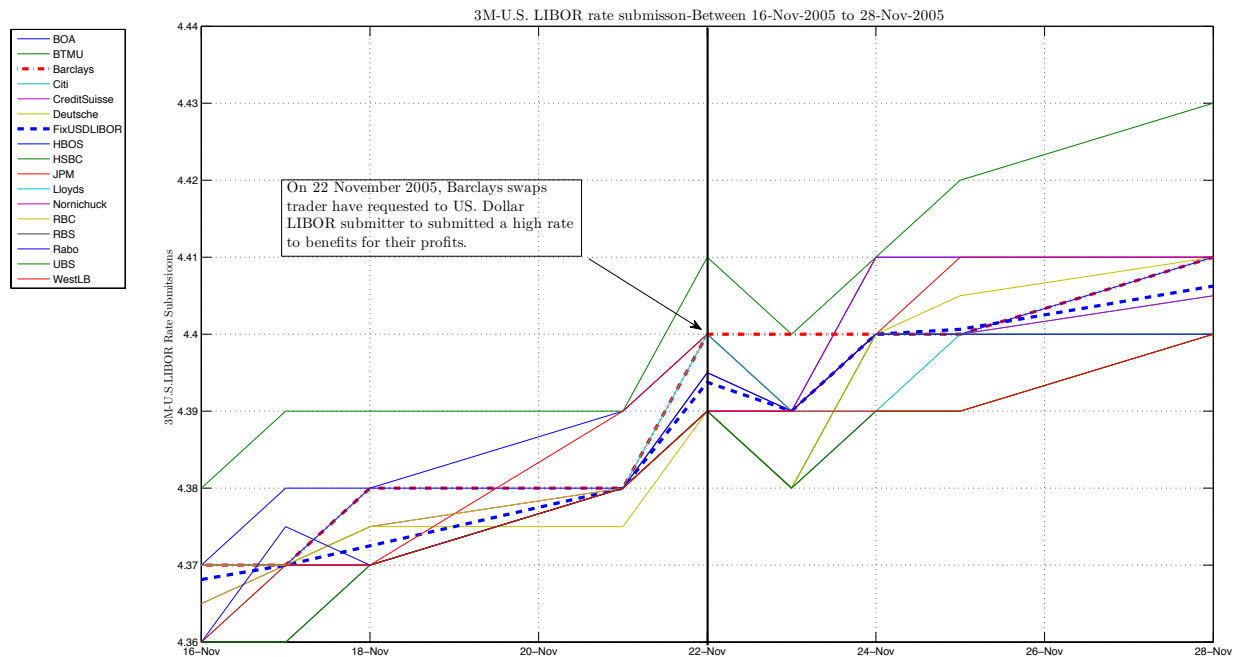


Figure 1: Barclays' 3M US Dollar LIBOR submission around November 22, 2005.

financed by the Commodity Futures Trading Commission (CFTC) and the Financial Services Authority (FSA). The documentation of manipulation of the LIBOR reference rate will be based on each bank's report. Overall, the reports cover a period of seven years from 2005 to 2012. We also focus on the reports primarily relating to the 3M US LIBOR submissions as this is the reference for our Eurodollar data.

The FSA (2012) report documents the earliest LIBOR manipulation event in 2005 when Barclays' derivatives employees made a total of 257 requests to Barclays LIBOR submitter to set the submission rate for LIBOR and Euribor. This was a routine request related to their derivatives trading positions and their own bonuses. Figure 1 shows an example graph showing that Barclays' submission was consistent with certain requests. For example, on November 22, 2005, Barclays' swap trader made a request in relation to Barclays' 3M US dollar LIBOR submission: "We have to get kicked out of the fixings tomorrow; we need a 4.41 fix in 3M (high fix)". The submitter sent a positive response to this request. Figure 1 shows the change in Barclays' submission compared to other panel banks' submission and the final 3M rate.

Table 2: Example of 3M-LIBOR submissions for six days between 2005–2008. The rates in bold are those that we averaged for the submission. The first three days, Nov 22, 2005, May 24, 2007 and Sep 4, 2007 all correspond to fixes that have been investigated and deemed suspicious. Sep 4, 2007 is a fix that occurs just after the sub-prime crisis, whilst the last two columns are samples of submissions before and after the Lehman Brothers collapse.

	Nov 22, 2005		May 24, 2007		Sep 4, 2007		Jun 18, 2008		Oct 15, 2008
<i>BTMU</i>	4.4	<i>Norinchuckin</i>	5.37	<i>Barclays</i>	5.21	<i>BTMU</i>	2.82	<i>Barclays</i>	5
<i>Barclays</i>	4.4	<i>HBOS</i>	5.365	<i>Deutsche</i>	5.21	<i>Norinchuckin</i>	2.82	<i>Cr. Suisse</i>	4.75
<i>BOA</i>	4.39	<i>BTMU</i>	5.36	<i>BTMU</i>	5.2	<i>Barclays</i>	2.81	<i>RBS</i>	4.75
<i>Citi</i>	4.39	<i>Barclays</i>	5.36	<i>BOA</i>	5.2	<i>HSBC</i>	2.81	<i>BTMU</i>	4.7
Cr. Suisse	4.39	BOA	5.36	Cr. Suisse	5.2	Rabobank	2.81	HBOS	4.65
<i>Deutsche</i>	4.39	<i>Citi</i>	5.36	<i>HBOS</i>	5.2	<i>WestLB</i>	2.81	<i>Deutsche</i>	4.6
<i>HBOS</i>	4.39	Cr. Suisse	5.36	<i>JPM</i>	5.2	<i>BOA</i>	2.8	<i>Norinchuckin</i>	4.6
<i>JPM</i>	4.39	<i>Deutsche</i>	5.36	<i>Lloyds</i>	5.2	<i>Citi</i>	2.8	<i>WestLB</i>	4.57
<i>Lloyds</i>	4.39	<i>HSBC</i>	5.36	<i>Norinchuckin</i>	5.2	Cr. Suisse	2.8	<i>UBS</i>	4.53
<i>Nornichuck</i>	4.39	<i>JPM</i>	5.36	<i>Rabobank</i>	5.2	<i>Deutsche</i>	2.8	<i>BOA</i>	4.5
<i>Rabo</i>	4.39	<i>Lloyds</i>	5.36	<i>RBC</i>	5.2	<i>HBOS</i>	2.8	<i>HSBC</i>	4.5
<i>UBS</i>	4.39	<i>Rabobank</i>	5.36	<i>RBS</i>	5.2	<i>Lloyds</i>	2.8	<i>Citi</i>	4.45
<i>WestLB</i>	4.39	<i>RBC</i>	5.36	<i>UBS</i>	5.2	<i>RBC</i>	2.8	<i>Lloyds</i>	4.45
<i>HSBC</i>	4.38	<i>UBS</i>	5.36	<i>WestLB</i>	5.2	<i>UBS</i>	2.8	<i>RBC</i>	4.45
<i>RBCc</i>	4.38	<i>WestLB</i>	5.36	<i>Citi</i>	5.14	<i>RBS</i>	2.795	<i>JPM</i>	4.1
<i>RBS</i>	4.38	<i>RBS</i>	5.35	<i>HSBC</i>	5.14	<i>JPM</i>	2.78	<i>Rabobank</i>	4.1
3M-LIBOR	4.39	3M-LIBOR	5.36	3M-LIBOR	5.2	3M-LIBOR	2.8025	3M-LIBOR	4.55

Figure 1 shows that Barclays' 3M US dollar LIBOR submission, which had been at the final benchmark rate, increased to a level above the benchmark rate on the day the swap trader requested a higher submission. It then remained high at the same level as the benchmark rate and finally increased again until the benchmark rate reached 4.41 as the swap trader had requested. Therefore, Barclays' submissions, on November 22, 2005 and later that week, were consistent with the request for a high 3M US Dollar LIBOR. The LIBOR investigation also shown that artificial rate had been tight in both high and low fixing, for instance, May 23, 2007 the Barclays swaps trader requested the submitter to submit a low LIBOR at 5.36 for tomorrow(May 24, 2007)as he sent email to submitter 'Pls., go for 5.36 again tomorrow, very long and would be hurt by higher setting...Thank.', then on May 23, 2007 Barclay submitted the 3M-USD LIBOR at 5.36 and the LIBOR was announced at 5.36 as their swap trader request. Another example was September 04, 2007 regarding to CFTC report the UBS swap trader explained to their manager why they requested a low LIBOR submission, so their submitter submitted a low rate at 5.2(see Table 2 for Barclay's LIBOR submission).

There is substantial evidence that Barclays' LIBOR fix was consistently manipulated during the volatile global conditions of the sub-prime crisis (circa 2007–2008), in order to

manage what it believed were inaccurate and negative public and media perceptions that it had a liquidity problem, by lowering its submissions – which were believed to be too low (see p. 3, CFTC, 2012). This activity continued after BNP Paribas House’s hedge funds had been frozen followed by the collapse of Northern Rock and Lehman Brothers. Finally, after Barclays raised £7.3 billion of capital from its Qatari shareholders, the CFTC began an investigation in the United States into the suspected “low-balling” of LIBOR submissions by global banks.⁴ In early December 2007, a Barclays’ employee alerted the U.K. Financial Services Authority (FSA) in which LIBOR rate setting at a dishonesty level was discussed (see p. 22, CFTC, 2012).

The analysis is continued by comparing the LIBOR submission rates with the benchmark rate before and after the collapse of Lehman Brothers. Figure 2 and Table 2 are examples of graphs and table showing that panel banks submitted rates that were believed to be abnormally low before the collapse of Lehman Brothers (June 18, 2008). When the end of Lehman Brothers made it clear to everybody that the crisis would last, the LIBOR rose sharply, whereas it had dropped heavily during the previous period to an almost “normal” level for an almost normal situation, but an abnormal level for a critical one (October 15, 2008).

In order to track the LIBOR manipulation, Timothy Geithner, President of the Federal Reserve Bank of New York at that time, sent a document containing a proposal for consultation to tackle the problem to Sir Mervyn King, the governor of the Bank of England.⁵ After receiving this document from the New York Fed, the FSA in London officially joined the international investigation, as a result of which the CFTC broadened its investigation. It decided to cooperate with the Securities and Exchange Commission (SEC) and began to communicate with multinational regulators, such as the European Commission (EC), the Swiss Competition Commission (ComCo), The Canada’s Competition Bureau and the

⁴P. Aldrick, Barclays: how the LIBOR manipulation unfolded, <http://www.telegraph.co.uk/finance/newsbysector/banksandfinance/9360469/Barclays-how-the-LIBOR-scandal-unfolded.html>, [accessed August 19, 2014].

⁵Timothy Geithner made recommendations on LIBOR to Sir Mervyn King (published online June 1, 2008), <http://data.newyorkfed.org/newsevents/news/markets/2012/LIBOR/June-1-2008LIBOR-recommendations.pdf>, [accessed August 19, 2014].

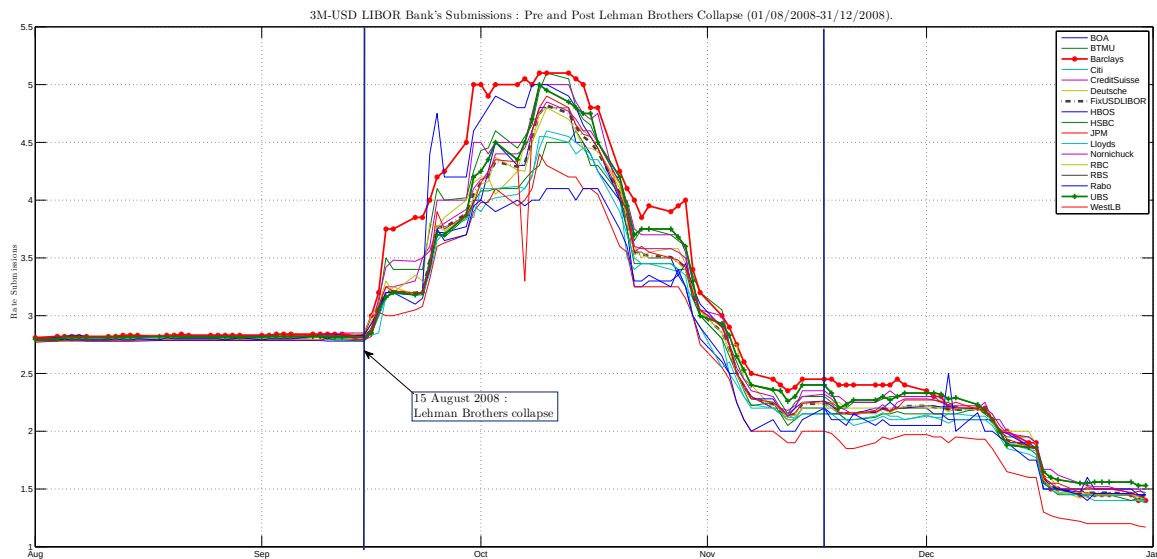


Figure 2: Barclays' 3M US Dollar LIBOR submission around Lehman Brothers collapse on 15 September 2008.

Japanese Regulator. A major effect of this cooperation was that the Swiss bank UBS agreed to end a tax dispute over US citizens and hand over 4,450 account details to assist the LIBOR investigation.

The FSA launched an investigation into Barclays to respond to allegations of LIBOR manipulation in June 2010, and in 2011, the FTC Capital and Charles Schwab Corporation filed a lawsuit against 12 major banks, including Barclays, RBS, HSBC and Lloyds, claiming that they had conspired to depress the LIBOR artificially.⁶ After the investigation, some of the major banks that were found to have manipulated the LIBOR rate, including Barclays and UBS, were fined by Commodity Futures Trading, the US Department of Justice, and the Financial Services Authority in the United Kingdom. Subsequently, RBS and ICAP were fined by US and UK regulators in 2013, and later the European Commission announced fines of six major banks for manipulating the LIBOR rate for the Japanese Yen in the period 2007–2010. After the US Department of Justice completed a criminal investigation into the LIBOR manipulation on June 27, 2012, Barclays was the first bank to be fined –

⁶M. Gilbert, G Finch and A. Worrachate, London banks seen rigging rates losing credibility with markets, Bloomberg Markets Magazine (published online Nov 22, 2011) <http://www.bloomberg.com/news/print/2011-11-23/london-banks-seen-rigging-rates-for-decades-losing-credibility-in-markets.html>, [accessed August 19, 2014].

\$200 million by the Commodity Futures Trading Commission (CFTC, 2012), \$160 million by the US Department of Justice and £144.5 million by the FSA (FSA, 2012) – for attempting to manipulate the world’s benchmarking borrowing rate, LIBOR.

From extensive textual analysis of the current settlements, it is evident that the 3M-LIBOR is the most commonly mentioned rate, followed by the 1M-LIBOR. A summary of the evidence of the abuse of the 3M-LIBOR and 1M-LIBOR is provided in Table 3, with an example of the communication content between traders and LIBOR submitters during the LIBOR manipulation.

Table 3: Timeline of the reports of LIBOR manipulation.

Year	Date	Maturity date	Bank	Inappropriate submissions following request by	Summary of email and conversations about LIBOR abuse	Action
2005	May 27	13/06/05	Barclays	Barclays’ derivatives traders requested to US Dollar LIBOR submitter.	“Who’s going to put my low fixing in?” The US Dollar LIBOR submitter replied “will be here if you have any requests for the fixing”.	Low setting
	Aug 2	19/09/05	Rabobank	The Rabobank US Dollar Desk Manager email to Senior Manager.	“Can you please set 6mL as high as possible?” Senior Manager reply on the next day: “Ok, leave it with us, will do our best.”	High setting
	Nov 22	19/12/05	Barclays	Barclays swaps traders request to US Dollar LIBOR submitters.	“We have to get kicked out of the fixings tomorrow, we need a 4.17 fix in 1m (low fix), We need a 4.41 fix in 3m (high fix).”	Low and High setting.
2006	Feb 1	13/03/06	Barclays	Barclays swaps traders request to US Dollar LIBOR submitters.	“We need 3m to stay low for the next 3 sets”.	Low setting
	Feb 3	13/03/06	Barclays	Barclays swaps traders request to US Dollar LIBOR submitters.	“Would love to get a high 1m, also a low 3m if possible, thanks”	Low and High setting.
	Feb 7	13/03/06	Barclays	Barclays’ derivatives traders requested to US Dollar LIBOR submitter.	“High 1m and high 3m if it’s possible please. Have v. large 3m coming up for the next 10 day or so.”	High setting
	Feb 15	13/03/06	Barclays	Barclays’ derivatives traders request to US Dollar LIBOR submitter.	Trader requested “Please go for [unchanged], or lower if poss”.	Low setting
	Mar 10	19/06/06	Barclays	Barclays’ derivatives traders request to US Dollar LIBOR submitter.	Two US Derivatives trades made email requests for a low three month US Dollar LIBOR submission for the coming Monday 13th.	Low setting
	Mar 16	19/06/06	Barclays	Barclays’ derivatives traders request to US Dollar LIBOR submitter.	Traders requested a high 1m and low 3m US Dollar LIBOR submission. Submitter responded “For you... anything. I am going to go 78 and 92.5.”	Low and High setting.
	Mar 27	19/06/06	Barclays	Barclays swaps traders request to US Dollar LIBOR submitters.	“We need low 1m and 3m LIBOR.”	Low setting
	Mar 31	19/06/06	Barclays	Barclays swaps traders request to US Dollar LIBOR submitters.	“We have another bix fixing tomorrow and with the market move I was hoping we could set the 1m and 3m LIBORs as high as possible.”	High setting

Continued on next page

Year	Date	Maturity date	Bank	Inappropriate submissions following request by	submissions request	Summary of email and conversations about LIBOR abuse	Action
	Apr 7	19/06/06	Barclays	Barclays' derivatives traders request to US Dollar LIBOR submitter.		Trader requested low 1m and 3m US Dollar LIBOR: "If it's not to low 1m and 3m would be nice, but please fell free to say 'no'. Coffees will be coming your way either way, just to say thank for your help in the past few weeks." A submitter responded "Done...for you big boy".	Low setting
	Jun 30	18/09/06	Rabobank	The Rabobank US Dollar trader email to US Dollar trader submitter.		"Morning mate, hope u got through yesterday ok. Usual request. Low 1s, high 3s if you can"	Low and High Setting
	Aug 6	18/09/06	Barclays	Barclays' derivatives traders requested to US Dollar LIBOR submitter.		Trader requested "Pls set 3m LIBOR as high as possible today?" Submitter responded "Sure 5.37 okay?"	High setting
	Sep 1	18/09/06	Rabobank	The Rabobank US Dollar trader email to US Dollar Trader submitter.		"High 3mth LIBOR for ur old mucker pls chief".	High setting
	Sep 13	18/09/06	Barclays	Barclays swaps traders request to US Dollar LIBOR submitters.		"Can we please keep the LIBOR fixing at 5.39 for the next few days. We do not want it to fix any higher than that."	Low setting
	Sep 15	18/12/06	Rabobank	The Rabobank US Dollar trader email to US Dollar Trader submitter.		"Usual favours, can you keep 3s LIBOR at 39 for the next few days pls mate, cheers".	High setting.
	Oct 13	18/12/06	Rabobank	The Rabobank US Dollar trader email to US Dollar Trader submitter.		"If you would be so kind and keep 3s low for monday's fix, Think it will come in at 37.5, would be a bit painful if was much higher."	Low setting.
	Dec 14	18/12/06	Barclays	Barclays swaps traders request to US Dollar LIBOR submitters.		"For Monday we are very long 3m cash here in NY and would like the setting to be as low as possible..'. Thanks."	Low setting
2007	Mar 16	18/06/07	Rabobank	The Rabobank US Dollar trader email to US Dollar Trader submitter.		"High 3mth LIBOR on Monday pls."	High setting.
	May 23	18/06/07	Barclays	Barclays swaps traders request to US Dollar LIBOR submitters.		"Pls., go for 5.36 LIBOR again tomorrow, very long and would be hurt by a higher setting....thanks."	Low setting
	Aug 3	17/09/07	Barclays	Barclays US Dollar traders request to US Dollar LIBOR submitters.		During the financial crisis, Barclays directed its US Dollar LIBOR submitters to lower their daily US Dollar LIBOR submissions in order to protect Barclays'reputation against market perceptions that Barclays had a liquidity problem based in part on its high LIBOR submissions.	Low setting
	Aug 10	17/09/07	UBS	The UBS US Dollar trader have requested to US Dollar LIBOR submitters.		Trader requested lower rate on 1 week and 2 week LIBOR US Dollar submission.	Low setting.
	Aug 13	17/09/07	Rabobank	The Rabobank US Dollar trader email to US Dollar Trader submitter.		"Gonna need a frickin high 6 mth fix tomorrow if ok with u, 5.42? I'll send a mail, write a post it not as well, it's 750 million 6mth fix! Could be risky!"	High setting.
	Aug 14	17/09/07	UBS	The UBS US Dollar trader request to US Dollar LIBOR submitters.		"my indications are deliberately on the low side ..."	Low setting.
	Sep 3 to 10	17/09/07	UBS	Conversation between the UBS US Dollar traders and US Dollar LIBOR submission manager (ALM Manager).		Traders explained to ALM Manager why UBS wanted to submit low LIBOR submission in most currencies.	Low setting.
	Sep 7	17/09/07	Rabobank	The Rabobank US Dollar trader email to US Dollar Trader submitter.		"If you can keep em up there that would save my arse a bit (til sep rolls off), cheers matey."	High setting.
	Oct 17	17/12/07	Rabobank	The Rabobank US Dollar trader email to US Dollar Trader submitter.		"A nice low 1 month for LIBOR for the rest of the week please matey. Cheers."	Low setting.

Continued on next page

Year	Date	Maturity date	Bank	Inappropriate submissions following request by	Summary of email and conversations about LIBOR abuse	Action
	Nov 5	17/12/07	UBS	Conversation between the UBS US Dollar traders and US Dollar LIBOR submission manager (ALM Manager).	Trader requested lower rate on LIBOR US Dollar submission.	Low setting.
	Nov 23	17/12/07	UBS	Conversation between the UBS US Dollar traders and US Dollar LIBOR submission manager (ALM Manager).	Trader requested lower rate on LIBOR US Dollar submission.	Low setting.
	Nov 28	17/12/07	Barclays	Barclays US Dollar traders have requested to US Dollar LIBOR submitters.	"LIBORs are not reflecting the true cost of money. I am going to set..., probably at the top of the range of rates set by LIBOR contributors...The true cost of money is anything from 5-15 basic points higher."	High setting
	Nov 29	17/12/07	Barclays	Barclays US Dollar LIBOR submitters convened a telephone discussion with the senior Barclays Treasury managers and the US Dollar LIBOR submitters.	The supervisor said "if the submitters submitted the rate for a particular tenor at 5.50, which was the rate they believed to be the appropriate submission", Barclays would be twenty basic points above the pack and it's going to causes shit storm.	Low setting
	Dec 4	17/12/07	Barclays	The Barclays senior US Dollar LIBOR submitter emails his supervisor.	[SIC] He stating that he submitted Barclays' 1m LIBOR at 5.30%, which was 4 basic points over the next highest submission and almost five basic points over the LIBOR FIXING.	High setting
2008	Feb 5	17/03/08	Barclays	The Barclays US Dollar LIBOR trader stated in a call to his manager.	Manager instructed trader to tell Barclays' submitter to keep LIBOR low.	Low setting
	Mar 3	17/03/08	Barclays	The Barclays senior US Dollar LIBOR submitter called to his colleague.	The manager told him that he should not be quite as aggressive in his submissions, meaning his submissions were too high.	Low setting
	Mar 17	16/06/08	Rabobank	The Rabobank US Dollar trader email to US Dollar Trader submitter.	"I know LIBORs are gonna come off today, I hv [SIC] quite a lot of fixings today, so the higher the better for me please."	High setting.
	Jun 17	16/06/08	UBS	Conversation between the UBS US Dollar traders and US Dollar LIBOR submission manager (ALM Manager).	US Dollar Trader-Submitter state that "we will start lowering over the next few days to get to more or less middle of the pack until future notice did you need me for any thing else?" ALM Manager replied "nope that was it, thx we should bring it down fast so we are in line by friday with the pack".	Low setting.

This is a synthesis of reports from the CFTC, FSA and FCA on the inappropriate US Dollar LIBOR submissions by employees of Barclays, Rabobank and UBS from 2005 to 2008. It focuses on the US dollar submission, particularly anything relating to 3-month submissions. The majority of reports on manipulation refer to the 1-month and 3-month LIBOR rates, predominantly the 3-month. This can be attributed to the fact that the Eurodollar Future is primarily a 3-month forward LIBOR rate, while the 1-month rates are commonly used for the floating rate leg of fixed to floating swaps. Since these are the products that have the highest notional valuations outstanding, they have the most potential for profitable informed trades.

4. Data Sources and Data Pre-processing

The data used in this research are introduced in this section, together with a preliminary processing of Eurodollar Futures data. As previously stated, Eurodollar futures contracts

are traded on the CME's Globex platform and CME pit (open-outcry) trades. Data for both electronic and open-outcry are directly obtained from the CME tapes for the period January 1, 1996 to January 1, 2014. Pit trades are quotes from CME by the ED code and GE for the Globex code; the amalgamated tapes are classified under the ED moniker. The volume ratio between the Globex and open-outcry is between three and four orders of magnitude over the sample, so separating the pit trades from the electronic trades currently appears to be less interesting. The CME tapes data for the 44 Eurodollar futures contracts available from the Thomson Reuter database via the Tick History system. We have conducted an analysis on the 4 monthly contracts, however the data is very sparse and the volumes are very small (up to 5 orders of magnitude for busy versus busy days) compared to the 40 quarterly contracts.

The data of the Eurodollar Futures trading are collected using the RICs (Reuter's Information Codes), which are quoted by financial institutions in the US Dollar exchanges. These 40 Eurodollar futures contracts are all Eurodollar transactions trading on the CME's Globex platform; they contain a maturity code, starting and ending dates, bid/ask price, bid/ask volume, trade prices and trade volume. The data pulled from the RIC information system, is a Computerized Trade Reconstruction (CTR) reported to the Commodity Futures Trading Commission (CFTC) on instruments in the tag range set, are given a specific code, i.e. EDU0, EDU1, ..., EDU9. The first two letters "ED" indicate Eurodollar futures, and the third letter "H, M, U, Z" indicates the delivery month of March, June, September and December respectively. The last character in the Eurodollar code, 0 to 9, represents the delivery year for 10-year future contracts from 2000 to 2009 or 2010 to 2019. For instance, in the case of EDH0, EDM1, EDU2 and EDZ3, their delivery months and years are March 2000, June 2001, September 2002 and December 2003 respectively (see Table 4).

Table 4 indicates the settlement year, the total volume of bid/asks, total number of bid/asks, total volume of trade and total number of trades for each Eurodollar futures contract. The data include all update-by-update inside quotes and trades for a total sample size of about 1 billion rows of prices and volumes in millisecond time-stamps and other trade information. This is believed to be an unprecedented microstructure data set as shown in

Figure 3: Number of bid/ask quotes and volume of quotes a day for EDH contracts EDH1, EDH2,..., EDH9 between 1996 and 2013.

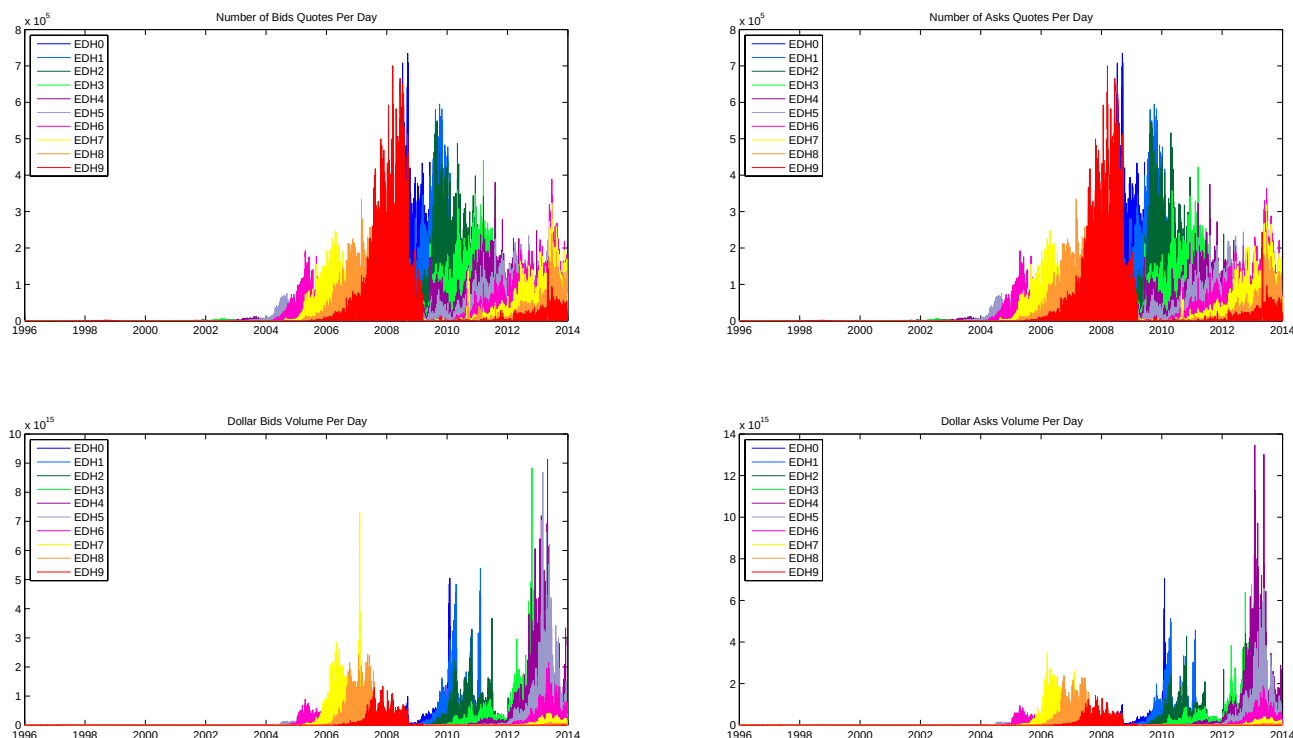


Table 4 and Figure 3.

Of the 40 Eurodollar future contracts, the largest bid volume is the EDZ3 contract at \$83.84 quadrillion. EDH4 has both the highest ask volume at \$92.80 quadrillion and the largest trade volume of \$120 trillion. Meanwhile, the biggest number of bids and asks belongs to EDU5 with approximately 25 million for both bids and asks, while EDH9 and EDZ9 has the largest number of trades with about 1.95 million for both series of contracts. The total value of bid/asks and trade from January 1, 1996 to January 1, 2014 was \$1.75 quintillion, \$1.76 quintillion and \$2.82 trillion, respectively. Furthermore, the total number of bid/asks and trades were approximately 831.36 million, 946.91 million and 63.83 million respectively.

Some information about the logistics of dealing with such a vast amount of data can now be provided. The raw data (uncompressed CSV files) comprised 489GB of best-bid, best offer and trade data, directly retrieved from the CME tapes. The raw tapes were streamed into a new format 'hdf5' which provided a high-integrity medium for this amount of data.

Table 4: Eurodollar Futures maturity codes, quotes volume and trades volume.

Maturity code	Delivery year	Bid volume (billion)	Ask volume (billion)	Trade volume (billion)	Bid up-dated (million)	Ask updates (million)	Trades (million)
EDH0	2000,2010, 2020	23.60	23.90	0.05	21.32	23.14	1.69
EDH1	2001,2011, 2012	52.70	52.10	0.06	22.34	23.04	1.51
EDH2	2002,2012, 2022	40.10	42.60	0.07	23.41	23.58	1.48
EDH3	2003,2013, 2023	56.10	56.60	0.05	24.12	24.49	1.31
EDH4	2004,2014	79.30	92.80	0.05	23.11	24.33	1.50
EDH5	2005,2015	57.20	58.80	0.06	21.61	25.40	1.81
EDH6	1996,2006,2016	31.10	30.40	0.07	18.33	25.97	1.82
EDH7	1997,2007,2017	42.00	40.10	0.06	13.62	23.03	1.38
EDH8	1998,2008,2018	34.40	33.90	0.11	12.23	21.25	1.48
EDH9	1999,2009,2019	20.10	20.10	0.09	18.20	22.97	1.95
EDM0	2000,2010, 2020	27.30	27.70	0.06	21.51	23.02	1.67
EDM1	2001,2011, 2012	56.60	57.60	0.06	22.78	23.42	1.49
EDM2	2002,2012, 2022	39.10	39.50	0.06	23.67	23.93	1.48
EDM3	2003,2013, 2023	61.90	66.10	0.04	23.35	23.87	1.30
EDM4	2004,2014	83.30	80.20	0.05	23.00	24.26	1.55
EDM5	2005,2015	45.20	48.30	0.08	20.87	25.56	1.92
EDM6	1996,2006,2016	28.20	28.10	0.08	17.09	25.52	1.76
EDM7	1997,2007,2017	40.00	40.00	0.07	12.40	22.11	1.31
EDM8	1998,2008,2018	31.30	30.90	0.10	13.69	21.55	1.55
EDM9	1999,2009,2019	17.00	17.00	0.08	19.56	23.29	1.93
EDU0	2000,2010, 2020	30.80	30.10	0.07	22.79	22.86	1.63
EDU1	2001,2011, 2021	52.80	67.20	0.02	23.47	23.34	1.49
EDU2	2002,2012, 2022	38.30	38.30	0.06	24.55	24.33	1.44
EDU3	2003,2013, 2023	77.30	74.90	0.05	23.94	23.84	1.39
EDU4	2004,2014	75.70	73.80	0.06	24.60	24.40	1.64
EDU5	2005,2015	39.50	40.70	0.08	25.73	25.71	1.93
EDU6	1996,2006,2016	31.20	31.00	0.08	25.00	24.98	1.60
EDU7	1997,2007,2017	40.60	39.60	0.09	21.35	21.30	1.32
EDU8	1998,2008,2018	27.60	29.20	0.01	22.05	21.88	1.61
EDU9	1999,2009,2019	15.90	16.30	0.07	19.56	23.29	1.93
EDZ0	2000,2010, 2020	42.80	43.70	0.07	21.32	23.14	1.69
EDZ1	2001,2011, 2021	46.80	46.00	0.07	22.34	23.04	1.51
EDZ2	2002,2012, 2021	44.80	45.50	0.06	23.41	23.58	1.48
EDZ3	2003,2013, 2023	83.80	82.10	0.06	24.12	24.49	1.31
EDZ4	2004,2014	67.00	70.00	0.07	23.11	24.33	1.50
EDZ5	2005,2015	36.20	36.10	0.09	21.61	25.40	1.81
EDZ6	1996,2006,2016	36.80	36.50	0.08	18.33	25.97	1.82
EDZ7	1997,2007,2017	38.10	37.90	0.12	13.62	23.03	1.38
EDZ8	1998,2008,2018	24.00	23.70	0.11	12.23	21.25	1.48
EDZ9	1999,2009,2019	18.10	18.30	0.06	18.20	22.97	1.95
Total		1,754.60	1,767.60	2.82	831.36	946.91	63.83

Eurodollar Futures maturity codes, quotes volume and trades volume from January 1, 1996 to January 1, 2014. The volume of bid/asks is reported in terms of 100 billion contracts. The trade volume is reported in millions of contracts.

Price and volume data were stored in a number format in order to reduce the amount of storage space as much as possible (e.g. prices were stored using IEEE single-precision numbers; volumes were stored at 16 bit integers). With compression, the raw 489GB was reduced to 40GB of compressed hdf5 data stored in separate files by maturity date, and then stored on a solid state drive. In the interests of speed, the data were replicated across three separate drives.

The PIN calculations were computed for each day by multiple instances of Matlab, on a large multiprocessor workstation with 0.5TB of RAM and virtual memory. The total computation time was around three weeks.

4.1. Data Cleaning Procedures

One of the first steps for this research is data collection to support all data analysis. Despite the fact that ED Trades and Quotes data is provided by a trustworthy data base, a precise data cleaning technique is still important to correcting data processing tasks. The cleaning of this data was carried out following steps S1-S4. These steps are applied with all trades and quotes data.

S1. Delete entries a bid, ask or trade price lower than 80 and NaN value.

S2. Delete entries a bid, ask or trade volume equal to zero and NaN value.

S3. Retain entries missing values and match with equivalent time vector for each variable.

S4. Match trades and quotes data by using Matlab function: `interp1` with 'nearest' method.

4.2. Eurodollar futures descriptive statistics

The data for our 40 Eurodollar future contracts is separated into two different timeframes, the first of which was from January 1, 1996 to July 31, 2007, and the second, from August 1, 2007 to January 1, 2014. The descriptive statistics for each ED futures contract in the two periods are shown in Tables 5 and 6.

In terms of the first period, the highest mean bid/ask spread belongs to EDU1, and EDZ0 has the highest mean trades returns. The smallest mean bid–ask spread belongs to EDH6,

and EDU6 has the lowest mean trade returns. As for the second period, EDM7 presents both the largest mean spread and mean returns, while EDZ9 shows the lowest mean spread, EDZ9 shows the lowest mean spread, and EDM9 presents the lowest mean return.

5. The model

5.1. Probability of Informed Trading – PIN

The PIN is a measure of asymmetric information on a trading event between informed and uninformed trading, which was developed on the theoretical framework of Easley and O'Hara (1992), although the original PIN approach was introduced by Easley et al. (1996). They produced a mixed discrete-time and continuous-time sequential model of the trading mechanism, in which trades emerge when three classifications of economic agents – namely market makers, informed traders and uninformed traders collaborate. Abad and Yagüe (2012), explain that on trading day, market maker is competitive or strategic trader, informed and uninformed traders are risk neutral with probability of profit or loss at 0.5. Also, informed and uninformed traders arrive in the market with their unique arrival rate with a view to a trading game between the liquidity provider and trader. Despite this, the PIN is not a direct measurement observation, but rather a parameter of a microstructure model which is estimated by the numerical maximization of the likelihood function. This is explained by Easley et al. (1996), as considering the likelihood of the arrival of three different types of information on a trading day: no news, good news, and bad news. This set of options for market agents can be summarized in a standard tree diagram representing the trading process. This diagram gives the structure of the trading process, where α is the probability of an information event, δ is the probability of a low signal, μ is the rate of informed trade arrivals, and ε is the rate of uninformed buy and sell rate arrivals.

The tree in 4 describes the trading process of Easley et al. (1996). Individual trading periods of a single risky asset on trading days are indexed $i = 1, 2, \dots, I$ and time is continuous within a trading day, and is indexed by $t \in [0, T]$. The competitive market makers are ready to buy or sell the asset at bid and ask posted price during the trading period. Information

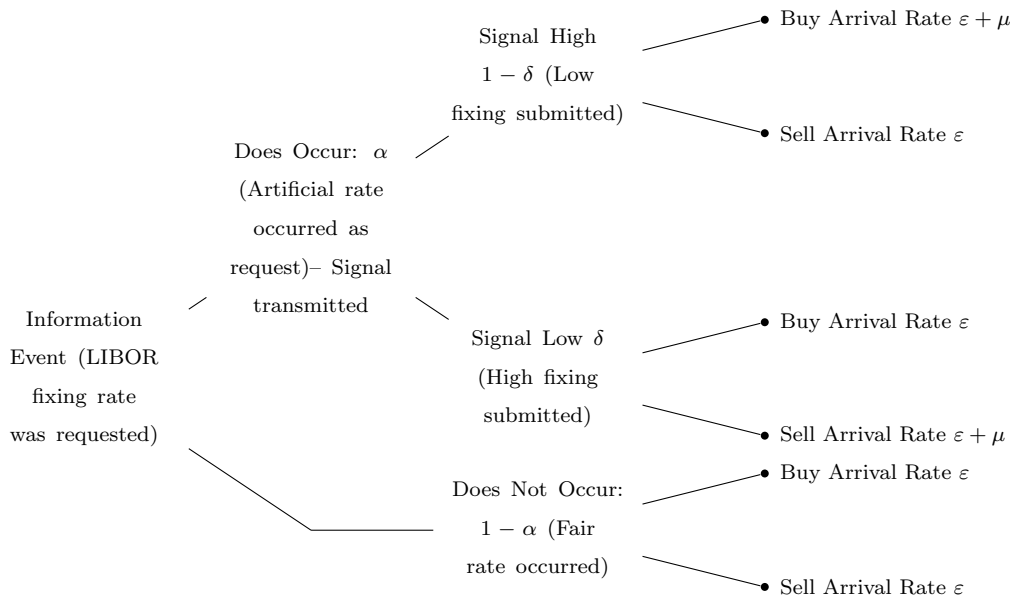
Table 5: The descriptive statistics for 40 Eurodollar Futures from 01/01/1996 to 31/07/2007 (BSP = basis point = $1e^{-5}$).

ED codes	Mean spread (BPS)	Median spread (BPS)	Max. spread	Min. spread	Std spread (BPS)	Skew spread	Kurt. spread	Mean re- turns $(\frac{1}{1000}(\text{BSP}))$	Median re- turns (BSP)	Mode re- turns	Max. re- turns	Min. re- turns	Std re- turns (BSP)	Skew re- turns	Kurt. re- turns
EDH0	1.5	1.1	1.1	0.05	0	3.3	42.31	2,718	9.4	0	0.01	-0.02	2.0	-10	2,682
EDH1	2.2	2.1	0.0	0.05	0	4.4	27.41	1,243	9.9	0	0.01	-0.02	1.8	-17	3,318
EDH2	2.2	0.5	0.0	0.07	0	8.5	23.64	806	8.1	0	0.01	-0.06	2.2	-166	42,329
EDH3	2.1	0.5	0.0	0.04	0	7.4	14.83	334	6.0	0	0.01	-0.06	1.8	-199	60,534
EDH4	1.7	1.0	0.5	0.11	0	4.8	37.79	6,385	6.2	0	0.01	-0.06	2.0	-224	70,601
EDH5	0.9	0.5	0.5	2.25	0	28.0	798.46	643,160	1.6	0	0.03	-0.03	0.9	-74	60,354
EDH6	0.6	0.5	0.5	0.06	0	1.0	83.01	18,832	-1.2	0	0.04	-0.04	0.9	-37	133,760
EDH7	0.7	0.5	0.5	0.06	0	2.8	67.07	6,078	-1.7	0	0.00	-0.03	0.6	-102	39,999
EDH8	0.7	0.5	0.5	0.06	0	1.2	134.67	41,985	0.9	0	0.01	-0.01	0.6	-9	2,470
EDH9	0.9	0.5	0.5	0.12	0	1.0	152.55	68,286	3.9	0	0.04	-0.04	1.6	-7	51,223
EDM0	1.6	1.6	0.0	0.05	0	1.9	51.65	6,883	10.5	0	0.01	-0.01	1.8	-2	1,432
EDM1	2.3	2.1	0.0	0.05	0	4.6	26.50	1,074	9.8	0	0.01	-0.04	2.1	-65	12,384
EDM2	1.7	0.5	0.0	0.05	0	5.5	20.31	690	7.6	0	0.01	-0.05	2.3	-124	29,914
EDM3	2.2	1.0	0.0	0.05	0	6.4	23.39	1,083	6.3	0	0.01	-0.05	1.8	-143	37,740
EDM4	1.4	0.5	0.5	0.09	0	3.4	37.07	5,656	5.0	0	0.01	-0.06	1.6	-235	82,308
EDM5	0.7	0.5	0.5	0.04	0	1.8	34.65	2,570	2.2	0	0.03	-0.03	0.8	-33	86,622
EDM6	0.6	0.5	0.5	0.04	0	1.2	128.12	25,789	1.5	0	0.04	-0.04	0.9	-51	142,750
EDM7	0.7	0.5	0.5	0.04	0	2.7	57.02	3,837	1.5	0	0.02	-0.02	0.7	-37	24,936
EDM8	1.0	0.5	0.5	0.06	0	1.7	84.40	13,153	1.5	0	0.00	-0.01	0.6	-9	1,794
EDM9	1.0	0.5	0.5	0.05	0	2.0	45.00	4,634	5.7	0	0.01	-0.03	1.3	-49	10,723
EDU0	1.8	1.6	0.0	0.03	0	2.5	31.97	1,890	10.4	0	0.01	-0.01	2.1	-4	1,523
EDU1	2.6	2.1	0.0	0.07	0	9.0	28.83	1,342	8.9	0	0.01	-0.04	2.1	-88	17,866
EDU2	1.4	0.5	0.0	0.05	0	4.5	29.47	1,520	6.5	0	2.23	-2.23	88.0	0	64,403
EDU3	2.4	1.0	0.5	0.02	0	6.7	8.89	131	6.7	0	0.01	-0.06	2.0	-202	60,072
EDU4	1.1	0.5	0.5	0.06	0	2.8	44.94	5,681	3.7	0	0.01	-0.05	1.2	-220	85,682
EDU5	0.6	0.5	0.5	0.03	0	1.2	64.37	7,310	1.2	0	0.04	-0.04	0.8	-20	115,010
EDU6	0.6	0.5	0.5	0.04	0	2.2	83.66	9,009	-1.7	0	0.04	-0.04	0.9	-41	149,020
EDU7	0.6	0.5	0.5	0.05	0	1.0	270.08	113,040	-0.1	0	0.01	-0.02	0.6	-40	12,697
EDU8	0.7	0.5	0.5	0.04	0	1.2	85.98	15,599	2.2	0	0.01	-0.01	0.7	-11	2,296
EDU9	1.1	1.1	0.5	0.05	0	2.5	37.43	2,714	6.9	0	0.01	-0.03	1.4	-56	11,566
EDZ0	2.0	1.6	0.0	0.06	0	3.4	1.34	2,288	12.2	0	0.01	-0.01	1.9	0	1,368
EDZ1	2.4	1.5	0.0	0.05	0	6.7	21.53	668	9.5	0	0.01	-0.06	2.5	-149	35,570
EDZ2	1.8	0.5	0.0	0.04	0	7.7	18.65	425	5.9	0	0.01	-0.05	1.9	-164	47,837
EDZ3	2.2	0.5	0.5	0.06	0	7.3	13.92	418	6.6	0	0.01	-0.06	1.9	-213	65,295
EDZ4	0.9	0.5	0.5	0.07	0	2.6	53.35	8,458	2.0	0	0.04	-0.04	1.2	-77	63,379
EDZ5	0.6	0.5	0.5	0.05	0	1.4	78.46	10,575	1.2	0	0.04	-0.04	0.8	-12	138,980
EDZ6	0.7	0.5	0.5	0.06	0	3.6	78.76	7,154	-1.6	0	0.01	-0.02	0.6	-74	29,202
EDZ7	0.7	0.5	0.5	0.16	0	1.2	385.68	433,280	0.5	0	0.01	-0.01	0.6	-6	2,095
EDZ8	0.8	0.5	0.5	0.03	0	1.9	42.25	2,436	0.3	0	0.01	-0.01	0.9	-10	4,836
EDZ9	1.4	1.1	1.1	0.06	0	3.5	30.23	1,619	9.1	0	0.01	-0.02	1.8	-16	2,878

Table 6: This table illustrates descriptive statistic for 40 Eurodollar Futures contracts from 01/08/2007 to 01/01/2014 (BSP = basis point = $1e^{-5}$).

ED codes	Mean spread (BPS)	Median spread (BPS)	Max. spread	Min. spread	Std spread (BPS)	Skew spread	Kurt. spread	Mean re- turns ($\frac{1}{1000}$ (BSP))	Median re- turns (BSP))	Mode re- turns	Max. re- turns	Min. re- turns	Std re- turns (BSP)	Skew re- turns	Kurt. re- turns
EDH0	1.0	0.5	0.5	0	1.9	131.61	70,698	0.3	0	0	0.01	-0.06	0.6	-610	607,830
EDH1	1.3	0.5	0.5	0	3.9	87.38	27,615	0.4	0	0	0.01	-0.05	0.6	-535	493,030
EDH2	1.3	0.5	0.5	0	3.8	93.19	29,520	0.5	0	0	0.00	-0.04	0.5	-376	302,630
EDH3	1.7	0.5	0.5	0	3.8	9.57	296	0.6	0	0	0.04	-0.04	0.7	-122	205,080
EDH4	1.4	0.5	0.5	0	6.6	38.25	4,723	4.2	0	0	0.02	-0.02	1.0	0	4,966
EDH5	1.6	0.5	0.5	0	1.6	136.37	30,339	4.5	0	0	0.01	0.00	0.6	0	297
EDH6	1.4	0.5	0.5	0	9.2	19.51	410	4.8	0	0	0.01	-0.01	0.9	0	206
EDH7	1.9	0.5	0.5	0	12.0	30.70	2,337	6.2	0	0	0.01	-0.01	1.1	0	257
EDH8	1.9	0.5	0.5	0	11.0	29.64	2,167	1.6	0	0	0.01	-0.03	0.9	-45	15,031
EDH9	1.1	0.5	0.5	0	9.8	64.45	4,384	0.4	0	0	0.01	-0.03	0.4	-206	147,460
EDM0	1.0	0.5	0.5	0	2.2	44.83	5,650	0.4	0	0	0.01	-0.04	0.5	-437	388,800
EDM1	1.5	0.5	0.5	0	5.1	45.22	8,734	0.4	0	0	0.01	-0.05	0.6	-546	500,910
EDM2	1.4	0.5	0.5	0	3.4	33.23	3,024	0.5	0	0	0.00	-0.03	0.4	-219	145,100
EDM3	1.6	0.5	0.5	0	4.2	16.02	557	0.6	0	0	0.00	-0.05	0.7	-277	190,220
EDM4	1.5	0.5	0.5	0	7.0	21.31	593	4.0	0	0	0.01	0.00	0.9	0	153
EDM5	1.3	0.5	0.5	0	9.6	44.04	11,161	2.0	0	0	0.01	-0.01	0.7	1	377
EDM6	1.4	0.5	0.5	0	9.5	19.05	395	5.0	0	0	0.14	-0.14	2.3	0	301,010
EDM7	4.5	0.5	0.5	0	34.9	132.00	17,439	6.8	0	0	0.01	-0.01	1.2	0	200
EDM8	1.5	0.5	0.5	0	19.0	899.14	1,072,400	1.0	0	0	0.01	-0.03	0.8	-67	27,438
EDM9	1.0	0.5	0.5	0	2.1	110.22	49,876	0.3	0	0	0.01	-0.05	0.5	-512	492,050
EDU0	1.1	0.5	0.5	0	1.2	27.63	2,217	0.4	0	0	0.01	-0.04	0.5	-430	377,590
EDU1	1.4	0.5	0.5	0	6.1	40.62	5,931	0.5	0	0	0.01	-0.04	0.5	-335	261,480
EDU2	1.5	0.5	0.5	0	27.0	4,774.40	23,235,000	0.5	0	0	0.00	-0.03	0.4	-271	190,970
EDU3	1.5	0.5	0.5	0	0.6	20.72	586	0.5	0	0	0.02	-0.05	1.0	-76	39,028
EDU4	1.6	0.5	0.5	0	9.6	23.24	1,508	4.2	0	0	0.00	0.00	0.8	0	172
EDU5	1.2	0.5	0.5	0	7.8	22.93	566	4.6	0	0	0.01	-0.01	0.7	0	984
EDU6	1.6	0.5	0.5	0	11.0	19.01	404	5.5	0	0	0.17	-0.17	2.8	0	311,730
EDU7	1.7	0.5	0.5	0	15.0	1,051.20	1,648,500	3.9	0	0	0.01	-0.01	1.1	0	213
EDU8	1.3	0.5	0.5	0	11.0	773.70	1,198,900	0.7	0	0	0.01	-0.02	0.6	-67	30,066
EDU9	1.0	0.5	0.5	0	1.7	135.07	89,802	0.3	0	0	0.01	-0.05	0.5	-556	552,290
EDZ0	1.3	0.5	0.5	0	4.3	74.98	19,688	0.4	0	0	0.01	-0.05	0.6	-594	571,520
EDZ1	1.5	0.5	0.5	0	6.5	32.37	3,584	0.5	0	0	0.01	-0.03	0.4	-286	211,740
EDZ2	1.5	0.5	0.5	0	3.4	20.63	1,921	0.5	0	0	0.00	-0.04	0.5	-348	264,280
EDZ3	1.3	0.5	0.5	0	5.9	27.31	887	0.5	0	0	0.00	-0.05	1.0	-87	43,136
EDZ4	1.5	0.5	0.5	0	9.4	24.15	1,569	4.3	0	0	0.01	-0.01	0.7	0	249
EDZ5	1.2	0.5	0.5	0	8.2	21.36	493	4.4	0	0	0.17	-0.17	2.3	0	493,600
EDZ6	1.8	0.5	0.5	0	12.0	15.60	275	5.7	0	0	0.01	-0.01	0.9	1	276
EDZ7	2.9	0.5	1.0	0	14.7	160.87	26,011	2.6	0	0	0.53	-0.53	9.3	0	318,150
EDZ8	1.3	0.5	0.5	0	12.0	473.28	613,670	0.5	0	0	0.01	-0.01	0.4	-25	10,243
EDZ9	0.9	0.5	0.5	0	2.1	195.95	80,831	0.3	0	0	0.01	-0.06	0.6	-624	635,860

Figure 4: Diagram 1. Tree diagram of the sequential trading progress (Easley et al., 1996).



Note: that when the fix is high, the price of the contract will decrease and when the fix is low the price will be higher, if the bank can guarantee than its fix will impact the LIBOR average determination.

events are independently distributed at the beginning of each trading day, and occur with probability α . However, informed traders will only trade on information event days; they will buy an asset if they receive a signal of good news with probability $1 - \delta$ and sell if they receive a signal of bad news with probability δ . As ε_b and ε_s are the selling and buying rate of uninformed traders, which are supposed to have the same intensity (0.5), therefore the probability of uninformed traders is $\varepsilon_s = \varepsilon_b = \varepsilon$. In this way, the set of parameters are reduced to only four: α, δ, μ and ε .

On a trading day, trade occurs from both uninformed traders and informed traders who trade on condition of a received signal. Both types of uninformed buyers and sellers arrive in the market at rate ε . Competitive informed traders who are risk neutral will arrive on the trading days on which the information events have occurred. If they receive a signal of good news, they will arrive to buy orders; conversely, they will submit sell orders if they receive a bad signal or bad news. The arrival rate for this process is μ . These arrival processes are assumed to be independent. Then, on good event days, the arrival rate for buy orders is

$\varepsilon + \mu$ and for sell orders is ε . On the other hand, if there is any bad news or bad signal on any trading day, the arrival rate of sell orders is $\varepsilon + \mu$ and buy orders arrive at a rate of μ . Finally, if there is no news or no signal on that day, only uninformed traders arrive for both buy and sell orders at arrival rate ε .

In each period the news arrival contains one of the three types of information. The market maker knows that there is some probability attached to each branch, and he knows the order arrival process for this branch. However, the market maker does not know which of the three branches has been naturally selected. Since he cannot directly observe which type of branch has been selected, Easley et al. (1996) assumes that the market maker is a Bayesian who uses the arrival of trade and the rate of trading to update his beliefs about the nature of the information event on the trading day. Let $P(t) = (P_n(t), P_b(t), P_g(t))$ be a liquidity provider's belief about the occurrence of information event "no news" (n), "bad news" (b), and "good news" (g) at time t . Then, his prior beliefs at time 0 is $P(t_0) = (1 - \alpha, \alpha\delta, \alpha(1 - \delta))$.

To determine quotes at time t , the market maker updates his prior belief on the condition of an arrival order of the relevant type. For instance, the bid at time t , $B(t)$, is the expected value of the asset conditional both on the history of the process prior to the arrival of order at time t and on the fact that someone wants to sell the asset. Let (s_t) denote the event that a sell order arrives at time t . Let $P_n(t|s_t)$ be the market maker's updated belief vector conditional on the history prior to time t and on the event that a sell order arrives at time t .

Following Bayes' rule, the market maker's posterior probability on no news at time t , if an order to sell arrives at t , is

$$P_n(t|s_t) = \frac{P_n(t) \cdot \varepsilon}{\varepsilon + P_b(t)\mu}, \quad (5.1)$$

the posterior probability on bad news is

$$P_b(t|s_t) = \frac{P_b(t) \cdot (\varepsilon + \mu)}{\varepsilon + P_b(t)\mu}, \quad (5.2)$$

and the posterior probability on good news is

$$P_g(t|s_t) = \frac{P_g(t) \cdot \varepsilon}{\varepsilon + P_b(t)\mu} . \quad (5.3)$$

Comparable to the case of buy orders (B_t) the posterior probability on good news at time t is

$$P_g(t|b_t) = \frac{P_g(t) \cdot (\varepsilon + \mu)}{\varepsilon + P_g(t)\mu} , \quad (5.4)$$

the posterior probability on bad news is

$$P_b(t|b_t) = \frac{P_b(t) \cdot \varepsilon}{\varepsilon + P_g(t)\mu} , \quad (5.5)$$

and the posterior probability on no news is

$$P_n(t|b_t) = \frac{P_n(t) \cdot \varepsilon}{\varepsilon + P_g(t)\mu} . \quad (5.6)$$

At the end of the trading on any day, the full information value of the asset is realized. If it is good news on trading day (i) the informed trader knows that the value of the asset at the end of the day is worth $\bar{S}_i$, similarly it is $\underline{S}_i$ if it is bad news on day i and the asset on day i is worth $S_i^* = \delta \underline{S}_i + (1 + \delta) \bar{S}_i$, (so, $\bar{S}_i > S_i^* > \underline{S}_i$) if there is no news at all.

To complete the bid or ask at time t , the liquidity provider updates his position on the condition of arrival order according to information type. At time t the expected value of the asset, conditional on the history of trade prior to time t , is

$$\mathbb{E}[S_i|t] = P_n(t)S^* + P_b(t)\underline{S}_i + P_g(t)\bar{S}_i , \quad (5.7)$$

where $S_i^* = \delta \underline{S}_i + (1 + \delta) \bar{S}_i$ is the prior expected value of the asset.

The bid is the expected value of the asset conditional on someone wanting to sell the

asset to a liquidity provider. Thus, it is

$$B(t) = \mathbb{E}[S_i|t] - \frac{\mu P_b(t)}{\varepsilon + \mu P_b(t)} (\mathbb{E}[S_i|t] - \underline{S}_i). \quad (5.8)$$

Similarly, the ask is the expected value of the asset conditional on someone wanting to buy the asset from a liquidity provider. Thus, it is

$$A(t) = \mathbb{E}[S_i|t] + \frac{\mu P_g(t)}{\varepsilon + \mu P_g(t)} (\bar{S}_i - \mathbb{E}[S_i|t]). \quad (5.9)$$

If there is no news on a trading day, so there are no informed traders ($\mu = 0$), then both bid and ask are equal to the expected value of the asset. Generally, both informed and uninformed traders will be in the market, so $A(t) > \mathbb{E}[S_i|t] > B(t)$.

The bid–ask spread at time t is denoted by $\Sigma(t) = A(t) - B(t)$. This spread is

$$\Sigma(t) = \frac{\mu P_g(t)}{\varepsilon + \mu P_g(t)} (\bar{S}_i - \mathbb{E}[S_i|t]) + \frac{\mu P_b(t)}{\varepsilon + \mu P_b(t)} (\mathbb{E}[S_i|t] - \underline{S}_i). \quad (5.10)$$

The first term in the bid–ask spread equation is a probability of buy order based on information and the second is the term for sells. The spread for the initial quotes in the period, Σ , has a particularly simple form in the natural case in which good and bad events are equally likely. That is, if $\delta = 1 - \delta$, then

$$\Sigma = \frac{\alpha\mu}{\alpha\mu + 2\varepsilon} (\bar{S}_i - \underline{S}_i). \quad (5.11)$$

The key component of this model is the probability that an order is from an informed trader, which is called the PIN:

$$PIN = \frac{\alpha\mu}{\alpha\mu + 2\varepsilon}, \quad (5.12)$$

where $\alpha\mu + 2\varepsilon$ is the arrival rate for all orders and $\alpha\mu$ is the arrival rate for information-based orders. Therefore, the PIN is a measure of the fraction of orders that arise from informed traders relative to the overall order flow, and the spread equation shows that it is the key

determinant of spreads. These equations clarify that liquidity providers need to correctly estimate the PIN in order to identify the optimal levels at which to enter the market. An unanticipated increase in PIN will result in losses to those liquidity providers who do not adjust their prices.

It is difficult to estimate the parameter vector $\theta = (\alpha, \delta, \varepsilon, \mu)$ because it cannot be directly observed in either the occurrence of information events or the associated arrival of uninformed and informed traders. In fact, in terms of measuring the daily arrival rate of sell (s_t) and buy (b_t) it is possible to infer these values using maximum likelihood and assuming that the trading process follows a Poisson process (Karyampas and Paiardini, 2011).

The Easley et al. (1996) PIN approach considers the likelihood of order arrivals on a day of known type. In this model, the likelihood of observing sell S and buy B orders on each type of information occurs on a no event day with probability $1 - \alpha$, a bad event day with probability $\alpha\delta$ and a good event day with probability $\alpha(1 - \delta)$. Thus, the likelihood is

$$\begin{aligned} L((B, S)|\theta) = & (1 - \alpha)e^{-\varepsilon_b} \frac{(\varepsilon_b)^B}{B!} e^{-\varepsilon_s} \frac{(\varepsilon_s)^S}{S!} \\ & + \alpha\delta e^{-\varepsilon_b} \frac{(\varepsilon_b)^B}{B!} e^{-\varepsilon_s + \mu} \frac{(\varepsilon_s + \mu)^S}{S!} \\ & + \alpha(1 - \delta)e^{-\varepsilon_b + \mu} \frac{(\varepsilon_b + \mu)^B}{B!} e^{-\varepsilon_s} \frac{(\varepsilon_s)^S}{S!}. \end{aligned} \quad (5.13)$$

Since only one type of information occurred on trading days, the maximum likelihood estimator of the information event parameters α and δ will be either 0 or 1. However, these parameters can be estimated from the daily numbers of buy and sells, assuming that the days are independent. The likelihood of observing the data $M = (B_i, S_i)_{i=1}^I$ across I days is just the product of the daily likelihood function

$$L(M|\theta) = \prod_{i=1}^I L(\theta|B_i, S_i). \quad (5.14)$$

The PIN estimates are computed by maximizing the parameter vector θ from any data set M , which is normally taken to be from the daily trades and quotes. For certain experiments

we also use 30, 60 and 90 day PIN estimates, these experiments are described in the following section.

Finally, we note that our PIN measure is calculating following this steps. First, after cleaning all data, trades classification is used to classified transactions by using Lee&Ready algorithm. Then, we use a maximum likelihood estimation with this classification to find the best value of $\aleph, \delta, \mu$ and ε . So, the PIN is computed following equation 5.12.

6. Analysis and Implications

We will analyze the PIN measure in the following sections using various rolling windows to record time series variation in the Eurodollar Futures PIN. Our estimated PIN averages over 0.5 for the majority of the sample and this is high relative to comparable equity market studies, see Table 7. However, our results coincide with those of Dhatt et al. (2012) on the CME Globex trades within our sample and Easley et al. (2012b) for the US Dollar Treasury note.

Table 7: Comparison between average PIN estimates across various studies and markets.

Author	Asset	Sample Period	Min-max	Mean
Easley et al. (1996)	NYSE – 90 stocks	Oct. 1, 1990 to Dec. 23, 1990	0.120–0.342	0.197
Idier and Nardelli (2011)	Euro overnight interbank rate, Money market	Dec. 2000 to Mar. 2008	0.200–0.580	0.480
Easley et al. (2011b)	E-mini S&P 500 (CME)	Jan. 1, 2008 to Oct. 30, 2010	0.205–0.830	0.393
Easley et al. (2012b)	E-mini S&P 500 (CME)	Jan. 1, 2008 to Oct. 30, 2010	0.205–0.830	0.393
	T-Note (CBOT)	Jan. 1, 2008 to Oct. 30, 2010	0.200–0.800	0.401
	EUR/USD(CME)	Jan. 1, 2008 to Oct. 30, 2010	0.150–0.780	0.327
	Brent Crude Oil (ICE)	Jan. 1, 2008 to Oct. 30, 2010	0.200–0.770	0.384
	Silver (COMEX)	Jan. 1, 2008 to Oct. 30, 2010	0.200–0.840	0.411
Abad and Yagüe (2012)	Spanish Stock Exchange – 15 stocks	Jan. 1, 2009 to Dec. 31, 2009	0.104–0.501	0.227
Dhatt et al. (2012)	Intra day trading – Eurodollar Futures (CME)	Jan. 3, 2005 to Dec. 29, 2006	0.760–0.970	0.880
Yan and Zhang (2014)	NYSE/AMEX stocks that have data in the ISSM and TAQ databases	Jan, 1, 1983 to Dec. 31, 2004	0.177–0.227	0.201
Our PIN	40 Eurodollar Future contracts	Jan. 1, 1996 to Jan 1, 2014	0.369–0.992	0.688

Although the PIN estimates range between 0.1 and 0.8 in the equities market, in the

derivatives market the range is higher, with minimum and maximum values of 0.20 and 0.97, respectively. Also, in the derivatives market the min–max spread is higher than in the equities market, particularly in the interest rate derivatives market. The highest mean of PIN is appear in Dhatt et al. (2012) with value of 0.880 ;however, the highest actual value of PIN is show in our work with the maximum value of 0.992. Interestingly, value of PIN in Eurodollar Futures is higher than equity market and always high as in our work the PIN is mostly higher than 0.5 and in Dhatt et al. (2012) the PIN varies between 0.76 and 0.97. Given the higher PIN that appears in the Eurodollar market in both these cases, it would be interesting to investigate this in the future.

6.1. Maximum Likelihood and PIN Estimation

The parameters of the PIN term structure model are estimated in this section. In fact, the trade process depends on four parameters – namely the probability of an information event; the probability of bad or good news; the arrival rate of informed traders; and the arrival rate of uninformed traders, presented as α, δ, μ and ε respectively. These parameters determine the probability of informed trading that is observed by order imbalance on information-based trading in the Eurodollar Futures market.

These four parameters of the trading process for each Eurodollar future series in a total of 40 data sets are estimated by maximizing the likelihood function, as described in Section 4. The mean estimate of the parameters and their standard deviation for each series are provided in Tables 8 and 9. Table 8 presents the results obtained from a preliminary analysis of the mean estimated parameters by Eurodollar futures series contracts.

Table 8: Mean value of PIN estimators and PIN for Eurodollar Futures market.

	α	δ	μ	ε	PIN
Mean	0.6444	0.3517	0.5604	0.5414	0.6882
Std dev.	(0.0533)	(0.0389)	(0.0087)	(0.0216)	(0.0068)

The mean value of the information event parameter α for Eurodollar contracts is 0.6444, which indicates that, on average, 3/5 of trading days are information driven. The second information parameter in the PIN approach is δ , which is the probability of bad news, so

that the signal is low (resp. high) during informed days with a probability δ (resp. $1 - \delta$). These results indicate that a high signal is observed with an estimated probability δ of 0.3517, and $1 - \delta$ is 0.6483. This means that the order is more likely to be buy than sell when an information event occurs. Thus, traders tend to believe that information in trading day-driven orders reveals an excess liquidity in demand on the Eurodollar futures market more than supply.

Thirdly, the estimated mean value of the rate of arrival of the informed trader μ is 0.5604, which suggests that Eurodollar traders tend to believe that observed orders based on an information-driven probability of 56.04% or just above half of the total orders that trade on the Eurodollar futures market come from informed traders.

Fourthly, the mean estimate ε is 0.54144, which indicates that the probability of liquidity trading is just above half. On informed days, the parameter ε represents the market liquidity that comes from uninformed traders, while on uninformed days, it coincides with the total liquidity available in the market because only uninformed traders have arrived in the market.

Having estimated the parameters $\alpha, \delta, \mu, \varepsilon$, these four parameters are used to calculate the PIN. According to equation (4.12) where the probability of information-based trading is $PIN = \frac{\alpha\mu}{\alpha\mu + 2\varepsilon}$, the PIN depends on the arrival rate of informed and uninformed traders and the probability of the occurrence of an information event.

The average value of the probability of informed trading or PIN on the Eurodollar Futures market for the whole Eurodollar futures series contracts was also calculated, and the mean value is reported in Table 8 as 0.6882. This suggests that investors who trade in the Eurodollar futures market face a 68.82% probability of trading with a counter-party who is better informed about the direction of the key rate.

The mean value of parameters for all 40 Eurodollar futures contracts was also estimated (see Table 9). The results shown in Table 9 illustrate separate descriptive statistics of the PIN value and PIN parameters on 40 Eurodollar futures series contracts from January 1, 1996 to January 1, 2014. These results are consistent with those of Easley et al. (1996), in that the parameters and the PIN measure are between 0 and 1. The interesting result in Table 9 is that the highest PIN value appears on EDH5 and EDM6 contracts, which have

the same value of 0.7, and EDH2 has the lowest PIN value of 0.677 (see Table 9).

Table 9: Maximum likelihood and PIN estimation.

	α	δ	μ	ε	PIN		α	δ	μ	ε	PIN
EDH0	0.585 (0.358)	0.407 (0.317)	0.551 (0.234)	0.540 (0.253)	0.679 (0.237)	EDM0	0.591 (0.359)	0.401 (0.318)	0.553 (0.237)	0.679 (0.257)	0.679 (0.242)
EDH1	0.601 (0.359)	0.396 (0.321)	0.549 (0.238)	0.542 (0.260)	0.679 (0.243)	EDM1	0.607 (0.353)	0.397 (0.321)	0.547 (0.235)	0.537 (0.262)	0.681 (0.241)
EDH2	0.599 (0.360)	0.395 (0.321)	0.546 (0.239)	0.540 (0.262)	0.677 (0.246)	EDM2	0.648 (0.344)	0.361 (0.319)	0.550 (0.233)	0.529 (0.269)	0.687 (0.239)
EDH3	0.665 (0.340)	0.339 (0.315)	0.556 (0.228)	0.520 (0.273)	0.694 (0.235)	EDM3	0.671 (0.338)	0.337 (0.316)	0.559 (0.227)	0.526 (0.272)	0.693 (0.232)
EDH4	0.685 (0.325)	0.319 (0.307)	0.564 (0.227)	0.521 (0.271)	0.696 (0.229)	EDM4	0.697 (0.316)	0.315 (0.305)	0.562 (0.225)	0.520 (0.271)	0.695 (0.228)
EDH5	0.716 (0.307)	0.296 (0.290)	0.572 (0.225)	0.533 (0.273)	0.700 (0.224)	EDM5	0.714 (0.314)	0.297 (0.203)	0.569 (0.229)	0.541 (0.274)	0.694 (0.229)
EDH6	0.713 (0.315)	0.291 (0.290)	0.575 (0.227)	0.545 (0.275)	0.695 (0.224)	EDM6	0.705 (0.319)	0.301 (0.300)	0.578 (0.229)	0.539 (0.279)	0.700 (0.228)
EDH7	0.665 (0.338)	0.334 (0.313)	0.572 (0.229)	0.538 (0.276)	0.694 (0.226)	EDM7	0.654 (0.347)	0.336 (0.313)	0.570 (0.227)	0.544 (0.274)	0.692 (0.226)
EDH8	0.628 (0.354)	0.365 (0.318)	0.559 (0.233)	0.550 (0.264)	0.680 (0.232)	EDM8	0.628 (0.354)	0.366 (0.318)	0.561 (0.232)	0.547 (0.261)	0.682 (0.23)
EDH9	0.599 (0.354)	0.390 (0.314)	0.556 (0.226)	0.541 (0.250)	0.683 (0.227)	EDM9	0.604 (0.354)	0.390 (0.317)	0.557 (0.228)	0.546 (0.252)	0.682 (0.229)

	α	δ	μ	ε	PIN		α	δ	μ	ε	PIN
EDU0	0.595 (0.3653)	0.404 (0.3246)	0.558 (0.2427)	0.554 (0.2603)	0.683 (0.2452)	EDZ0	0.598 (0.3643)	0.401 (0.3234)	0.550 (0.2404)	0.547 (0.2613)	0.678 (0.2453)
EDU1	0.605 (0.353)	0.393 (0.3162)	0.552 (0.2358)	0.536 (0.2606)	0.686 (0.2402)	EDZ1	0.623 (0.3457)	0.387 (0.3173)	0.548 (0.2362)	0.533 (0.2651)	0.684 (0.2424)
EDU2	0.653 (0.3438)	0.357 (0.3186)	0.552 (0.2368)	0.527 (0.2724)	0.687 (0.2442)	EDZ2	0.659 (0.3435)	0.348 (0.3181)	0.557 (0.2275)	0.523 (0.2717)	0.694 (0.233)
EDU3	0.673 (0.3336)	0.332 (0.3129)	0.557 (0.2268)	0.524 (0.2674)	0.691 (0.2321)	EDZ3	0.673 (0.3332)	0.333 (0.3126)	0.559 (0.2279)	0.523 (0.2682)	0.693 (0.2308)
EDU4	0.702 (0.3155)	0.309 (0.3059)	0.564 (0.225)	0.527 (0.2729)	0.696 (0.2281)	EDZ4	0.712 (0.3101)	0.300 (0.3029)	0.566 (0.2263)	0.534 (0.2727)	0.694 (0.2276)
EDU5	0.718 (0.3127)	0.295 (0.3059)	0.570 (0.2305)	0.548 (0.2725)	0.690 (0.2298)	EDZ5	0.722 (0.3106)	0.292 (0.3043)	0.576 (0.2287)	0.546 (0.2761)	0.697 (0.2251)
EDU6	0.692 (0.3289)	0.307 (0.3071)	0.574 (0.2282)	0.540 (0.2769)	0.696 (0.2255)	EDZ6	0.681 (0.3295)	0.318 (0.3059)	0.568 (0.2366)	0.541 (0.2766)	0.690 (0.2317)
EDU7	0.452 (0.3521)	0.348 (0.3174)	0.569 (0.2307)	0.543 (0.2734)	0.691 (0.2304)	EDZ7	0.639 (0.3494)	0.353 (0.3151)	0.566 (0.2303)	0.548 (0.2671)	0.686 (0.2299)
EDU8	0.619 (0.3541)	0.371 (0.315)	0.563 (0.2277)	0.550 (0.2582)	0.685 (0.2245)	EDZ8	0.615 (0.3482)	0.380 (0.3117)	0.557 (0.2248)	0.548 (0.2509)	0.681 (0.2232)
EDU9	0.595 (0.3539)	0.399 (0.3146)	0.552 (0.2303)	0.542 (0.2497)	0.679 (0.2323)	EDZ9	0.581 (0.3609)	0.410 (0.3167)	0.552 (0.2271)	0.542 (0.252)	0.681 (0.2308)

The table presents the mean and standard deviations of parameters of the PIN model, and the mean value of PIN for 40 Eurodollar futures series contract. The parameter α represents the probability of an information event, δ is the probability of a low signal, μ is the rate of informed trade arrival, and ε is the rate of uninformed buy and sell trade arrivals. PIN is the probability of informed trading.

6.2. PIN and Eurodollar Futures

The next step is to analyse the way in which some events, such as turmoil in the financial market or maturity effects, which have taken place in the Eurodollar futures market may

have affected the trading strategies, and also the probability of informed trading. In order to particularly assess the evolution of informed trading over time, the model parameters were estimated on samples using 78 overlapping days. Then the historical PIN was plotted separately into 10 sub-plots to illustrate the pattern of the PIN value. Each sub-plot contained four different ED contracts with an expiry date within the same years; for instance, the first sub-plot included EDH0, EDM0, EDU0 and EDZ0. These four ED contracts were due to expire in 2000 and 2010. The historical PIN is presented in Figure 5. Surprisingly, a similar trend of PIN appears in all 10 sub-plots. Furthermore, some landmarks of the historical PIN are indicated to assess whether some turning points in the trend of the PIN can be associated with major events that took place in the Eurodollar market, as shown in Figure 5.

Overall, the PIN estimates for the majority of Eurodollar futures contracts were high and widely fluctuated between 1996 and 2000; however a decreasing trend can be observed from 2002 until around 2005 for the majority of ED contracts and this was reversed to an upward trend between 2005 and 2006. Interestingly, this increasing trend after 2006 is before the sub-prime crisis. The PIN remained steady at around 0.6 from 2010 but was still below the historical PIN from the first period (1996–2000) (see Figure 5).

We can see from Figure 6 that four particular events in the four periods appear to have exerted some influence on the historical development of the PIN. The first period that influenced the PIN was between 1996 and 2000, when the PIN widely and highly fluctuated. This can possibly be related to the fact that, in this period, the Eurodollar Futures market was not popular or traded less frequently than today. There were very few quotes and trade volumes in that period of time – for example, in the EDM series (Figure 7). As a result, the Eurodollar market was uncomplicated and it was easy for informed traders to manipulate it by modifying their trading behavior to reflect a different degree of PIN. After this period, a decreasing trend of the PIN could be observed after 2002 and one possible explanation for this is that the number of quotes has increased rapidly, from less than 100 per day to more than 10,000 per day and around 50 trades per day have increased to more than 1,000 per day. Therefore, these results of a decreasing PIN could be affected by more-frequent trading

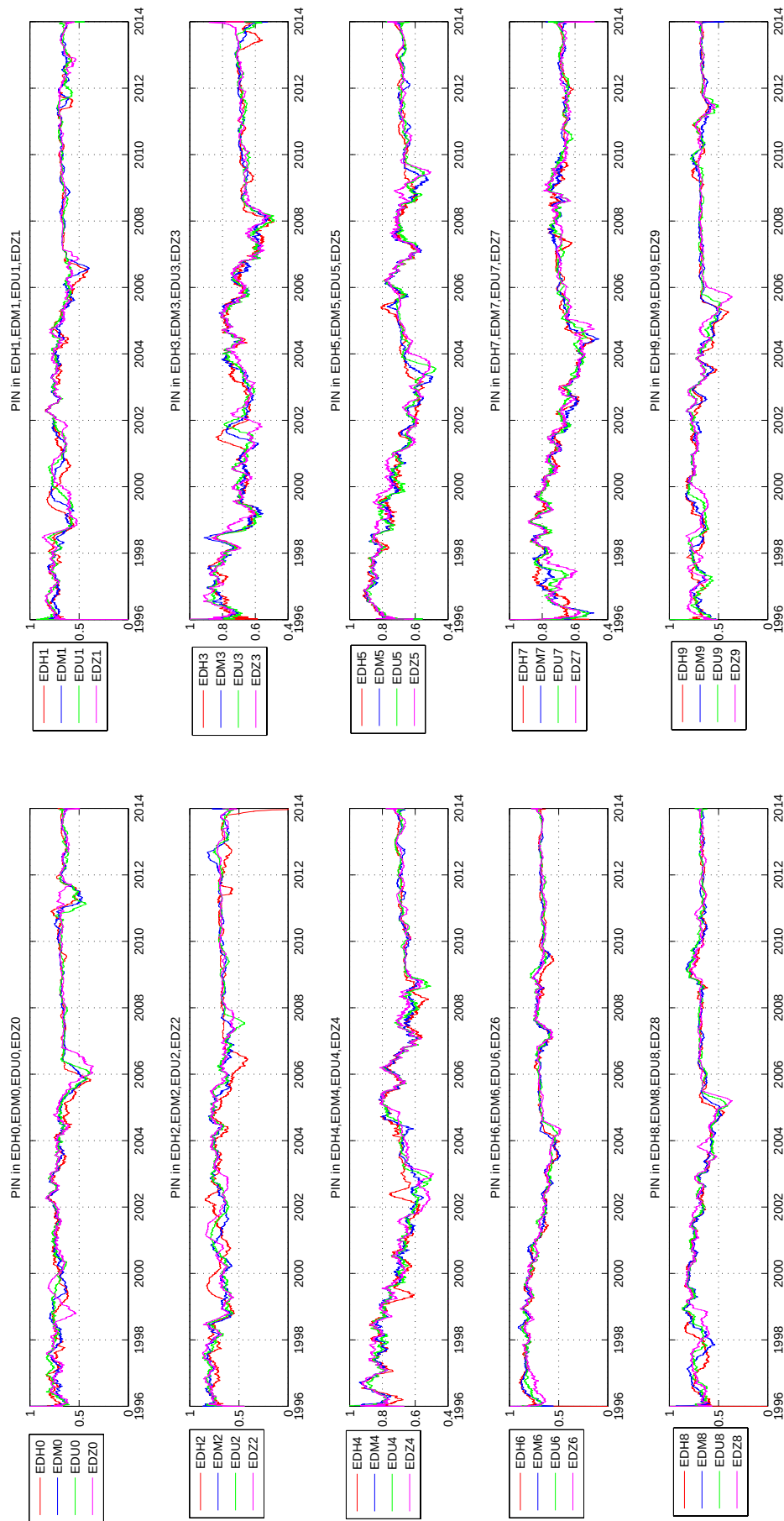


Figure 5: Historical evolution of the PIN. The historical PIN is separated into 10 groups, each group including four ED contracts that have an expiry date within the same years.

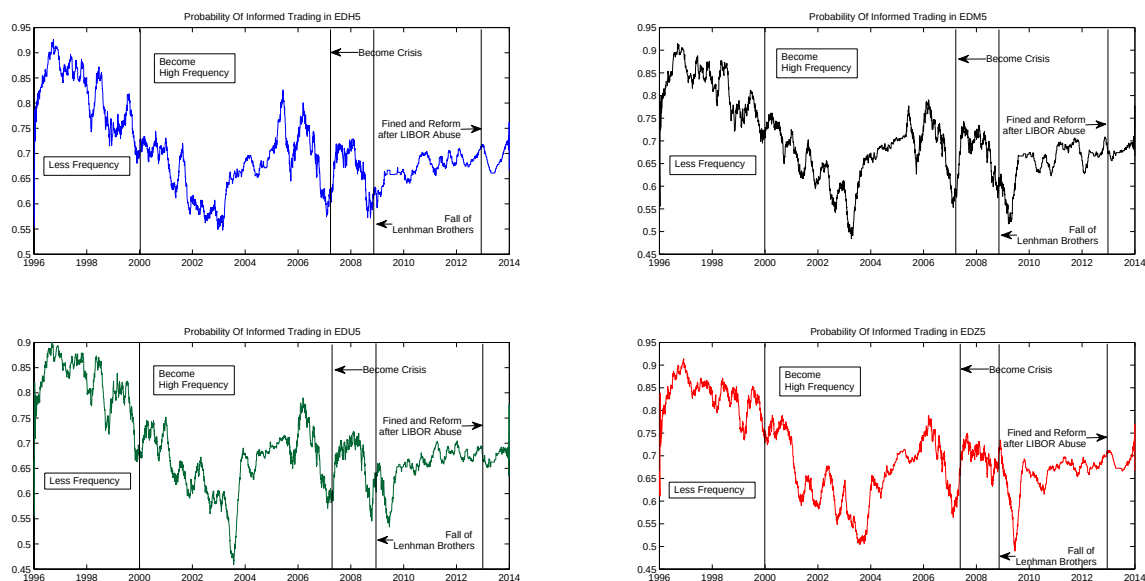


Figure 6: Historical PIN on EDH5 EDM5 EDU5 EDZ5.

This figure shows the historical value of PIN for four ED contracts (EDH5, EDM5, EDU5, EDZ5) with a timeline period that affects the change of PIN value. We plot the historical of PIN from 1996 to 2014, also we show the timeline of events that affect the PIN: first, the less-frequent trading period; second, increased frequency of trade after 2000; third, during the financial crisis; finally, after regulators fined the banks that were involved in LIBOR manipulation.

in futures contracts.

The third period of historical events that affected the value of the PIN was the market turmoil. The sub-prime crisis mainly affected the money market and used the market to reduce the liquidity in the financial market. The PIN sharply decreased at the beginning of the financial turmoil, approximately in mid-2007, after which it PIN changed rapidly because of some spiking in most Eurodollar contracts. This was possibly affected by the reduced liquidity at the beginning of the crisis, and subsequently the panel bank tried to maintain the key interest rate in the LIBOR submission to protect the reputation of the banks, as explained in the next section.

The final event that affected the historical PIN occurred in 2012. This was when the CFTA and the FSA announced their order to Barclays Bank, UBS, RBS and Rabobank to pay a penalty for manipulating the LIBOR. The effect of the announcement was visible in the marked decrease of the PIN in 2012 and it became more stabilized in most ED futures

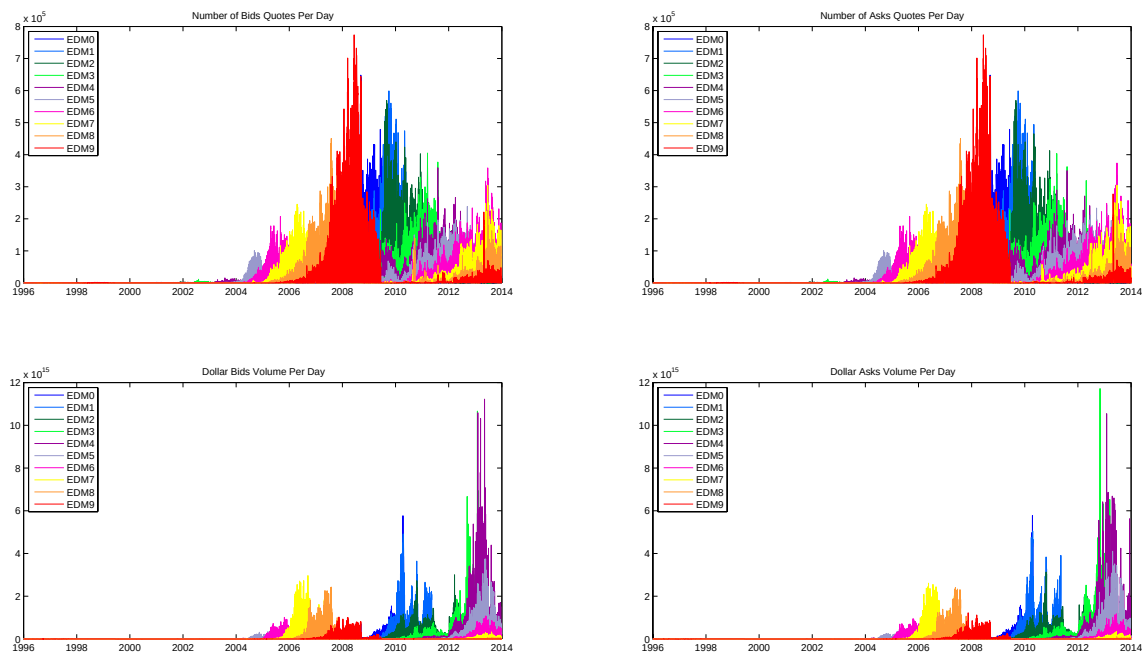


Figure 7: Example of number of quotes and volume of quotes per day for 10 EDM contracts. The number of quotes and trade were very low or less frequent from the opening in 1996 until 2000 with a significant increase after 2000 which reached a peak between 2008 and 2010, then a decreasing trend started from 2010. This trend occurred for all 40 Eurodollar Futures.

contracts after 2013.

6.3. PIN and LIBOR Manipulation

The following reviews the anecdotal trends in the estimated PIN coefficients for the various contracts relative to the specific events catalogued in Table 3 versus the longer run patterns observed in the data over the 1996 to 2014 sample period. This section begins by comparing the historical PIN with specific events related to the LIBOR manipulation records. Since the LIBOR abuse was revealed by the US Department of Justice, FSA and CFTC investigations, we make the conjecture that individual events had an impact on the PIN in the Eurodollar futures market as the ED trading is based on this key rate. A closer look at the way in which the LIBOR investigation caused changes in the PIN begins with a document analysis using the Commodity Futures Trading Commission (CFTC) document when the CFTC investigated some of the major banks, such as Barclays and RBS, regarding dishonest or manipulative submissions of the LIBOR rate. As previously noted, two epochs appear to be important in this context, first the pre-crisis attempts to generate competitive

advantage directly in the banks proprietary interest rate trading positions and second after the crisis as a systematic attempt to reduce the perception of increased borrowing costs as part of a beauty contest designed to ‘reassure’ investors and regulators.

According to the CFTC settlement document, Barclays’ traders had attempted to manipulate the US Dollar LIBOR from at least mid-2005 to the autumn of 2007, and sporadically thereafter until 2009. For instance, on November 28, 2007, Barclays’s employees, including the bank’s senior Treasury managers, submitted a US Dollar LIBOR rate that was higher than the actual rate and LIBOR abuse occurred on the following day and into early December 2007.⁷

This activity is investigated in this study from the PIN value of the EDH8 contract, which had the closest expiry date series to this date. The PIN increased from November 24, 2007, when its value was 0.54 and reached a peak of 0.715 on December 15, 2007. Figure 8 illustrates the manipulation that began on November 28, 2007. This increasing value of the PIN is consistent with the CFTC document; for example, on November 28, 29 and 30, 2007 the PIN values were 0.64, 0.65 and 0.66 respectively, and reached a peak of around 0.715 on December 15, 2007 as Barclays’ US Dollar LIBOR traders asked the submitter to keep the rate high.

There were many other instances of LIBOR abuse that can be illustrated by the movement of the PIN. The final notice from the Financial Services Authority (FSA) issued to Barclays Bank plc stated that at least 14 of the bank’s interest rate derivatives traders had requested the LIBOR submitter to fix the LIBOR rate. However, this was not always done only to improve their trading position, but also sometimes to protect the bank’s reputation. For instance, on February 5, 2008, Barclays’ US Dollar Derivatives manager instructed traders to keep the LIBOR rate lower than the real rate to make the bank look healthier than usual during the financial turmoil.⁸ It is interesting to see how this activity is reflected in the ED PIN to determine if the level of informed trading that occurred in the Eurodollar

⁷CFTC, Order instituting proceedings pursuant to sections 6(c) and 6(d) of the Commodity Exchange Act, as amended, making findings and imposing remedial sanctions; In the Matter of Barclays plc, pp. 20–22.

⁸Financial Services Authority, Final Notice To Barclays Bank plc, p. 11.

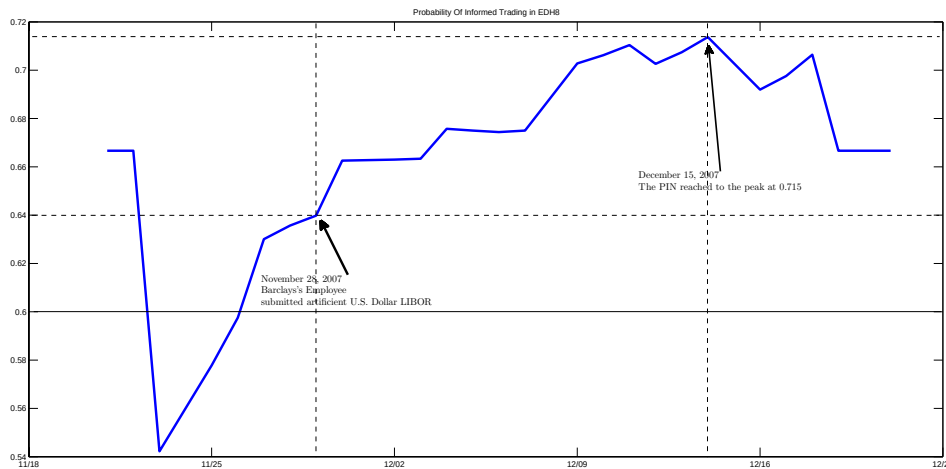


Figure 8: Probability of informed trading between November 28 and December 15, 2007. This figure shows the development of the PIN during the LIBOR abuse between November 28 and December 15, 2007. According to the CFTC document, Barclays' traders attempted to manipulate the US Dollar LIBOR on November 28, 2007, when Barclays' employees, including senior Barclays Treasury managers, submitted a higher US Dollar LIBOR rate than the actual rate.

market during this period changed substantially. Prima facie evidence indicates that the LIBOR abuse appears to have possibly exerted some influence on the historical development of the PIN. The PINs of EDM8, EDU8, EDZ8 and EDH9 are used to indicate the LIBOR setting from February 5 to 11, 2008. Of course, the limitation of this study is that that we cannot identify the specific trades undertaken by the specific banks as this information is not available publicly.

Figure 9 illustrates the historical PIN during this period. The PIN indicator shows an increasing trend from around 0.68 on February 3 2008 to just around 0.70 on February 5, 2008 reaching a peak of around 0.76–0.78 on February 11, 2008. This figure illustrates that the PIN was still in an upward trend until February 5, 2008, and the increase shown in this figure is possibly evidence of the abuse of Eurodollar Futures.

Barclays was not the only major high street bank involved in manipulating the LIBOR. Rabobank (Coöperative Central Raiffeisen Boerenleenbank B.A.), the Dutch multinational banking and financial services company headquartered in Utrecht was also guilty of this 'misdemeanour' and fined accordingly. According to the CFTC settlement report, CFTC (2013), the Rabobank LIBOR submitter submitted preferential rates in an attempt to manipulate the US Dollar LIBOR to benefit its trading position. In fact, Rabobank employees frequently

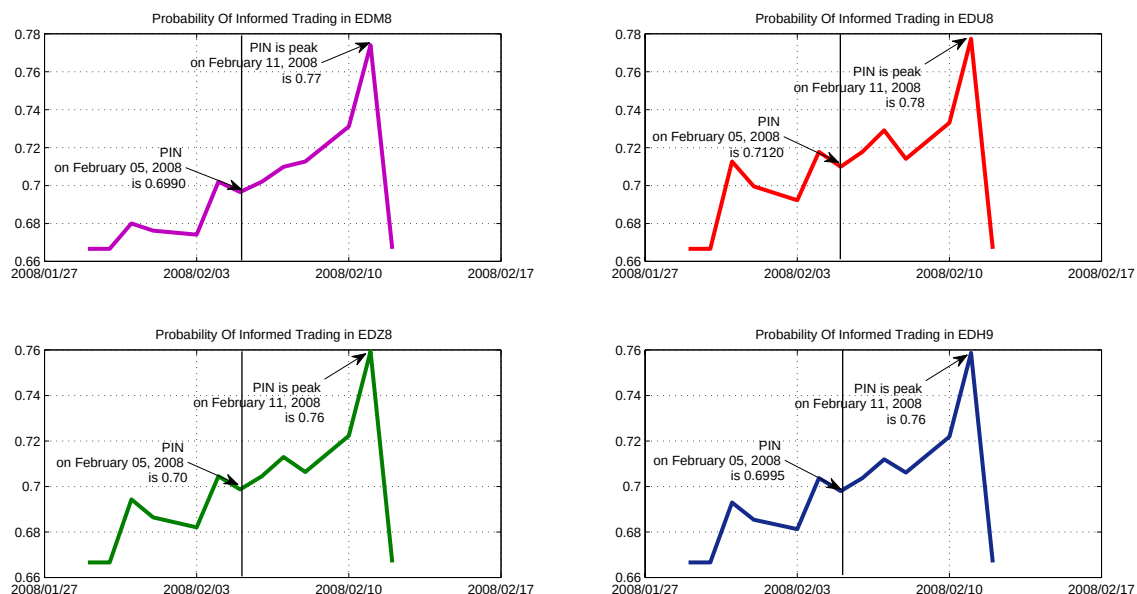


Figure 9: Probability of informed trading between February 5 and 11, 2008. This figure illustrates the development of the PINs of EDM8, EDU8, EDZ8 and EDH9 between February 3 and 10, 2008. On February 5, 2008 Barclays' Derivatives manager instructed traders to keep the LIBOR rate low. There appears to be an increasing trend of the PINs during this period and a drop after the event date.

attempted to manipulate the US Dollar LIBOR over a period from at least mid-2005 to at least late 2008. There are many examples of submitters being requested by traders to fix the LIBOR; for example, on September 07, 2007, Rabobank's senior US Dollar trader emailed the US Dollar submitter to keep the 3M US Dollar LIBOR high for the rest of the week.⁹ The fact that this period was characterized by an increasing trend of the historical PIN is evidence that the Eurodollar futures market was affected by the setting of the LIBOR. This manipulation influence on the EDU7 contract, which expired on September 19, 2007 or just about two weeks after the event, is used to investigate this incident by matching it with the US Dollar LIBOR. The results indicate that, on September 07, 2007 the PIN had increased from 0.68 on September 6 to 0.7 and it was kept higher than 0.70 for the remainder of the week (see Figure 10). The PIN reached a peak of 0.80 on September 17 and dropped to 0.695 on the following day. The historical PIN for this period could provide some evidence

⁹CFTC, Order instituting proceedings pursuant to detection(c) and 6(d) of the commodity exchange act, as amended, making findings and imposing remedial sanctions in the matter of Coöperative Central Raiffeisen Boerenleenbank B.A., p. 10.

that the Eurodollar market had been manipulated by informed traders; moreover, another result is that the value of the PIN during this period was higher than average for EDU7 (0.6855).

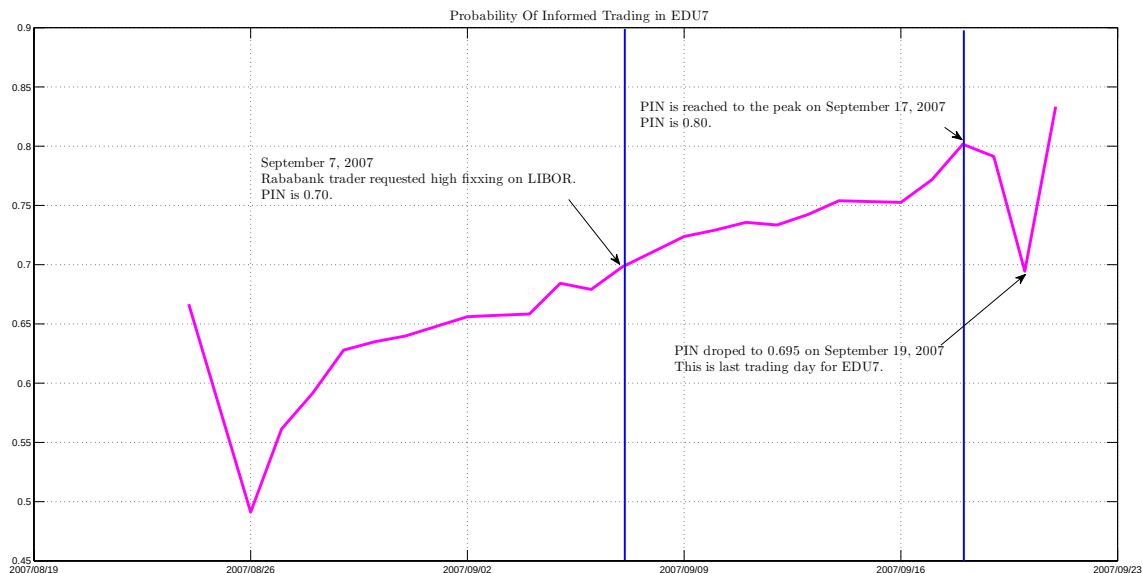


Figure 10: Probability of informed trading between September 7 and September 22, 2007. This figure presents the value of the PIN of EDU7, which reacted to the LIBOR abuse between September 7, 2007 and September 22, 2007 when Rabobank's US Dollar LIBOR traders asked the submitter to keep the LIBOR high for the rest of the week. As shown in Figures 5 and 6, the trend of the PIN also increased and remained high for the rest of that week. Since this trend could have been affected by the LIBOR abuse of Rabobank's employees, it is probably evidence to prove that the LIBOR was manipulated during this period.

Most of the LIBOR abuse activity that affected the Eurodollar futures trading occurred before and during the sub-prime crisis; however, one major event that had an impact on the LIBOR and Eurodollar Futures trading occurred much later, on February 28, 2012 when the US Department of Justice was conducting a criminal investigation into the LIBOR manipulation. Later that year, Barclays was the first bank to be fined \$200 million by the CFTC, \$160 million by the US Department of Justice, and £59.5 million by the UK's FSA for attempting to manipulate the key rate. Therefore, the Eurodollar Futures trading during this period is also investigated in this paper by simulating the PINs of EDH2, EDM2, EDU2 and EDZ2 contracts to compare them before, during and after 2012, when the LIBOR was manipulated. The action taken by the CFTC and FSA was a warning to traders to be wary of submitting an honest LIBOR rate and it should have reduced the instances of informed trading in the Eurodollar market. According to the investigation in this paper, there was

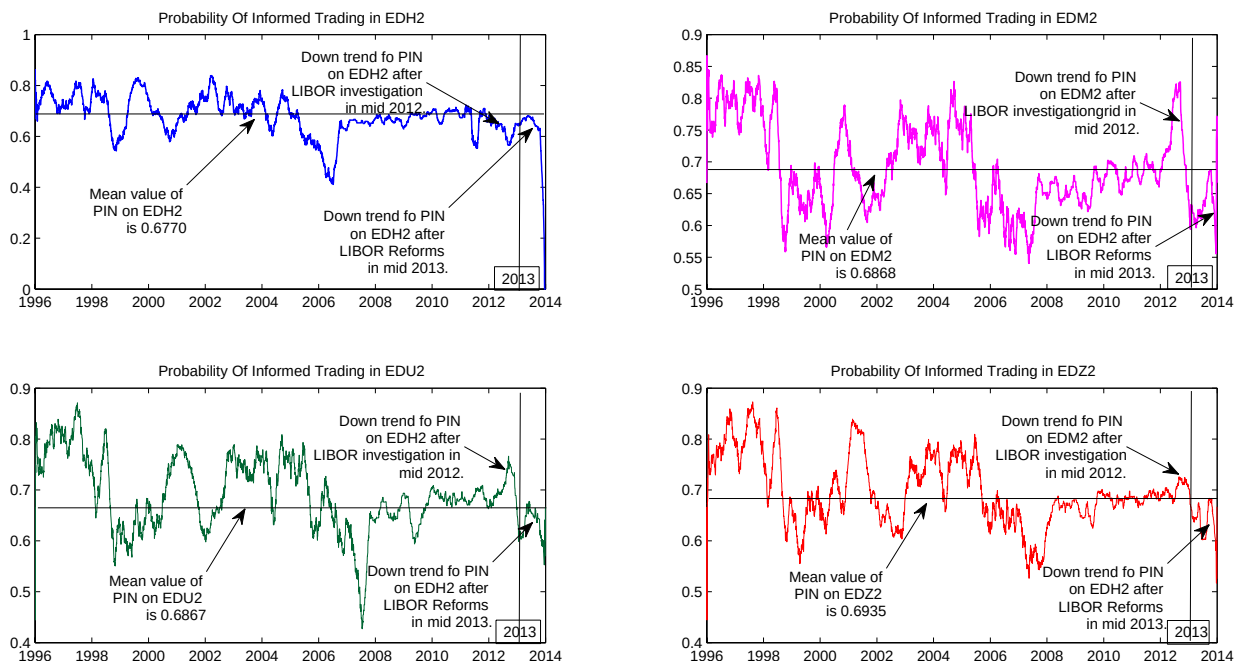


Figure 11: The probability of informed trading after LIBOR criminal investigation and reform. This figure demonstrates the modification of the PIN in two periods, the first of which was after the PIN investigation and the announcement of fines imposed on Barclays, RBS, UBS and Rabo bank by the CFTC. There was a decreasing trend of the PIN in the first quarter of 2012, after the investigation of all ED contracts, followed by an upward trend from the last quarter of 2012 until mid-2013. The second period was after the LIBOR reform in mid-2013, when another downward trend of PIN appeared, probably as a result of the new law to protect the LIBOR from manipulation.

significant drop in the PINs of these four contracts in mid-2012 (see Figure 11), especially EDM2, which expired in June 2012 with a maturity date around three months after the regulators announcement.

The EDM2 PIN dropped by 24% from around 0.84 to 0.60 from the beginning of 2012 to the end of that year. The decrease of the PIN would have been affected by the investigation, lawsuit and fines for LIBOR manipulation by the CTFC and the FSA, which could have influenced some informed traders. This influence is illustrated by the significant change in the historical PIN in 2012 with the downward trend after the LIBOR investigation (see Figure 11). Furthermore, after the LIBOR Regulation reform in July 2013, the PIN of every contract significantly dropped from around 0.68 to around 0.55 in the second half that year.

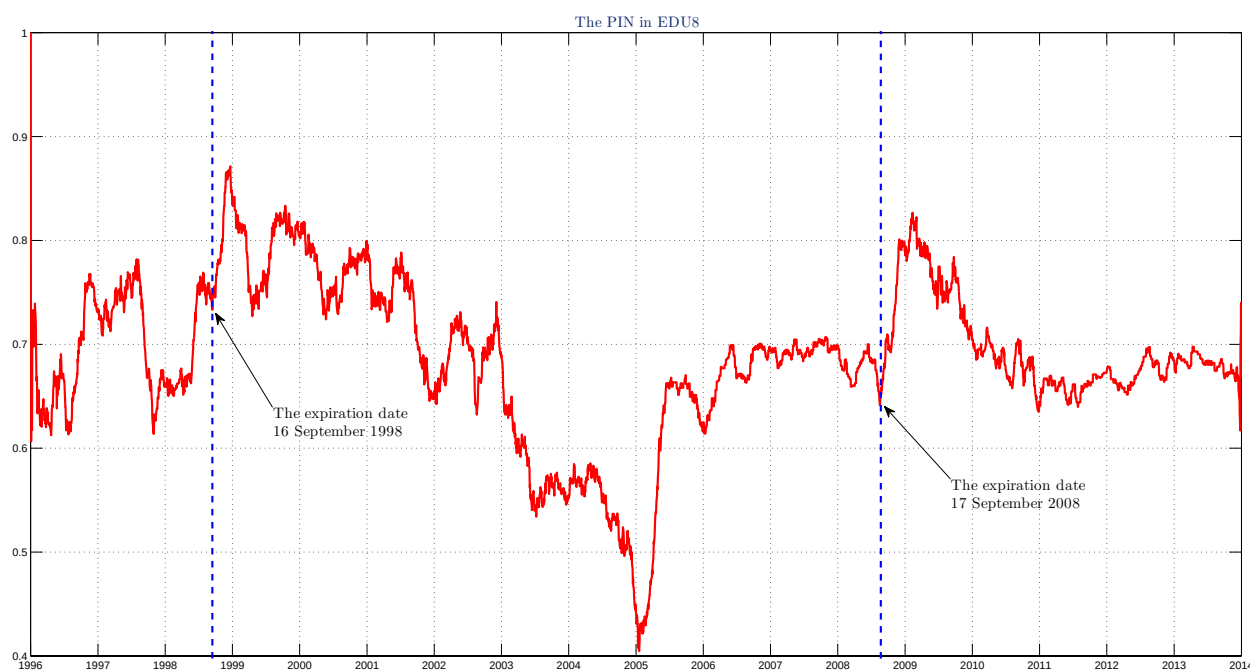
An interesting observation is that after the reform of the LIBOR fixing system and the exposure of bad practice, the PIN computed for the various Eurodollar contracts exhibits

a steady decline with lower than the average value of 0.68 after 2013. However, given the degree of fluctuation in the PIN through time and across contracts, further analysis is needed to determine the persistence of this trend, see Figure 11).

6.4. PIN and the Maturity Date Effect

This section contains an illustration of the PIN before and after expiration to investigate the relationship between the PIN and the maturity date. The evolution of the PIN is plotted with the expiration date of all 40 Eurodollar contracts to illustrate how the PIN reacted before and after the maturity date. The most striking result to emerge from this figure is that, after the spiking of the PIN before the expiry date, there was an aggressive decline until the last trading day, and finally, another increase appears when the Eurodollar futures begin trading again on the day after the expiry date (see Figure 12). This pattern of the PIN appears in all 40 Eurodollar series contracts.

Figure 12: The historical PIN of EDU8 with the expiry date: the PIN was dropping rapidly before the last trading day, but increased when EDU8 began trading again on the following day. This pattern is repeated in all the Eurodollar contracts.



The PINs for individual Eurodollar futures contracts are divided into two periods, the first of which is between January 1, 1996 and December 31, 2007, while the second is between

January 1, 2008 and December 31, 2013. The average value of the daily PINs was constructed for 60, 30, 20 and 10 days for fixed time windows around categories of identified events. The first identified sub-event was based on the regulatory activity around the current catalogue documenting LIBOR manipulation from the CFTC and the FSA, from Table 3 while the second was based on contract maturity dates by year (see Table 10).

In terms of the first sub-events, the average value of the PINs was not significantly different between 60, 30, 20 and 10 days for fixed time windows around regulatory based events. The smallest PINs appeared 10 days before the identified event for both the pre-2008 and post-2008 sample, whereas the highest PIN appeared 10 days after the event. Furthermore, in the pre-2008 period, the PINs slightly increased from the event date and reached a peak 60 days after the event with an average value of 0.7069.

As for the second sub-event, which was around the maturity date, the average PIN slightly increased from 60 days before the maturity date, and then the PIN dropped on the last trading day and increased again after the expiry date. Interestingly, the average PIN was identical on the maturity date, before and after 2008. Although there were slight differences in the evolution of the PIN for both regulatory-based events and around the maturity date, these differences were not found to be statistically significant (see Table 10).

7. Conclusions

We have undertaken an empirical microstructure analysis of the Eurodollar futures market, based around a rolling window adaptation of the model of Easley and O'Hara (1992) and Easley et al. (1996). This is probably the first attempt to apply a simple sequential trade model to analyse the reaction between the probability of informed trading and market abuse based on manipulating the LIBOR for highly active derivatives markets, such as Eurodollar futures, or LIBOR-based derivatives traded on the CME's Globex platform. However, the difference between equity and derivatives markets should be considered.

Overall, the PIN appears to have dropped around the time of the sub-prime crisis (circa 2007) and then bounced back afterwards and sharply dropped again with the collapse of Lehman Brothers and then bounced back once again indicating that systematic market

Table 10: Comparison of the results of the mean value and standard deviation.

Day before the event	Mean value of PIN for LIBOR abuse post-2008	Mean value of PIN for LIBOR abuse pre-2008	PIN before/after maturity date post-2008	PIN before/after maturity date pre-2008
-60	0.6752 (0.0089)	0.6938 (0.0196)	0.6933 (0.0324)	0.6926 (0.0223)
-30	0.6724 (0.0089)	0.6946 (0.0237)	0.6975 (0.0400)	0.6933 (0.0278)
-20	0.6717 (0.0088)	0.6957 (0.0260)	0.6999 (0.0433)	0.6926 (0.0313)
-10	0.6731 (0.0081)	0.6967 (0.0287)	0.7006 (0.0459)	0.6930 (0.0354)
0	0.6732 (0.0117)	0.6968 (0.0325)	0.6552 (0.4656)	0.6695 (0.0368)
10	0.6738 (0.0053)	0.6995 (0.0324)	0.7066 (0.0448)	0.6978 (0.0420)
20	0.6686 (0.0053)	0.7020 (0.0318)	0.7097 (0.0477)	0.6979 (0.0466)
30	0.6671 (0.0043)	0.7043 (0.0322)	0.7116 (0.0501)	0.6970 (0.0488)
60	0.6663 (0.0017)	0.7069 (0.0350)	0.7157 (0.0550)	0.6891 (0.0530)

Comparison of the results of the mean value and standard deviation of the PIN 60, 30, 20 and 10 days before and after the events of LIBOR manipulation according to the CFTC and FSA documents, and also before and after the maturity date in two periods: from 1996 to the end of 2007 and from 2008 to the end of 2013.

uncertainty has played a role in information asymmetry. The evolution of the PIN appears to vary systematically with maturity, since it dropped around the last trading day prior to maturity and then recovered when the Eurodollar futures resumed trading the following day. Significantly, the result of the PIN in this paper is always higher for the Eurodollar futures market than those recorded in the equity market (circa 0.4 to 0.8 versus 0.1 to 0.5; see Easley et al., 1996; Aslan et al., 2011; Abad and Yagüe, 2012), however other PIN studies of interest rate derivatives have found similarly high, if not higher PIN measurements, in particular Dhatt et al. (2012) find the PIN for the Eurodollar market to be substantially higher on average than that detected in the equity and FX markets.

In summary, in terms of illustrating the presence of informed trading in the Eurodollar futures market, the initial results are somewhat inconclusive, when taken as a whole. We find that in the lead up to some of the more blatant attempts at manipulating the LIBOR fix the PIN provides a good early warning signal. However, the average adjustments to the PIN relative to the dates recorded in the current catalogue of documentation from the CFTC and the FSA on LIBOR manipulation was not statistically significantly different relative to both the persistent variation relative to maturity and the general long-run variation in the PIN.

This may be because the PIN approach was unsuitable or it was possibly used incorrectly for this type of data, or the LIBOR manipulation only had a minor effect. However, we leave full investigation of this issue to future work as more granularity emerges from the current round of court cases provides more direct evidence of channels of informed trading (thus allowing us to better the cases).

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